

**GLASGOW CALEDONIAN UNIVERSITY**

School of Science and Engineering

**Data-Driven Condition Monitoring and Performance  
Analysis of Industrial Gate Valves**

*MSc Applied Data Science in Engineering*

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## Abstract

Industrial gate valves are critical components in high-pressure fluid systems, where performance degradation can lead to leakage, inefficiencies, and significant operational risk. This dissertation presents a data-driven exploratory analysis framework for assessing gate valve condition using experimental qualification test data provided by TechnipFMC, a global subsea and surface technology company.

Two distinct datasets were analyzed. The first comprises low-frequency cyclic pressure (leakage) test data from three valve pressure classes: 5,000 psi (5k), 10,000 psi (10k), and 15,000 psi (15k), each subjected to 500 open-close cycles at rated pressure. The second dataset consists of high-frequency vibration and magnetic sensor data from a dedicated HF limit test, recorded at 400 Hz over 92 complete valve actuation cycles across three test days.

For the cyclic leakage tests, a cycle-by-cycle KPI framework extracted peak hydraulic pressure, differential pressure, average, and minimum flow rate. The 5k valve exhibited clear pressure collapse and seat leakage failure from cycle 408; the 10k valve showed a deliberate test condition step-change at cycle 55 but passed qualification; and the 15k valve demonstrated stable performance throughout 500 cycles.

For the HF vibration test, valve position was derived from the magnetometer Z-axis (MagZ) signal. Stroke duration KPIs were extracted for all 92 cycles using EVENTS markers, revealing a progressive increase in both opening and closing stroke duration from Day 1 (opening mean 168s, closing mean 50s) to Day 3 (opening mean 210s, closing mean 80s). Vibration analysis using RMS, kurtosis, PSD, and STFT revealed that early cycles exhibited highly impulsive mechanical behaviour (kurtosis >200), consistent with a component bedding-in period, transitioning to broadband sustained vibration by cycle 3.

The study demonstrates that integrating cycle-based KPI analysis with multi-parameter vibration characterization provides substantially richer diagnostic information than a conventional binary pass/fail qualification outcome. The findings are consistent with TechnipFMC's internal Valve Signature Health Model framework. Future work should extend the analysis to all 92 HF cycles, incorporate actuator pressure signatures, and investigate automated anomaly classification.

## Acknowledgements

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## List of Abbreviations

| Abbreviation       | Definition                                        |
|--------------------|---------------------------------------------------|
| AccX / AccY / AccZ | Tri-axial Accelerometer Channels (X, Y, Z)        |
| API                | American Petroleum Institute                      |
| APDSCE             | Applied Data Science in Engineering (MSc Program) |
| CSV                | Comma-Separated Values                            |
| dB                 | Decibels                                          |
| FFT                | Fast Fourier Transform                            |
| GCU                | Glasgow Caledonian University                     |
| HF                 | High-Frequency                                    |
| HPU                | Hydraulic Power Unit                              |
| Hz                 | Hertz (cycles per second)                         |
| IMU                | Inertial Measurement Unit                         |
| ISO                | International Organization for Standardization    |
| KPI                | Key Performance Indicator                         |
| MagZ               | Magnetometer Z-axis (valve position proxy)        |
| MATLAB             | Matrix Laboratory (MathWorks)                     |
| PdM                | Predictive Maintenance                            |
| PSD                | Power Spectral Density                            |
| RMS                | Root Mean Square                                  |
| SDP                | Symmetrized Dot Pattern                           |
| STFT               | Short-Time Fourier Transform                      |
| TechnipFMC         | Technip and FMC Technologies (merged entity)      |
| TOC                | Table of Contents                                 |

# Chapter 1: Introduction

## 1.1 Background to the Study

Gate valves are widely used as isolation devices in high-pressure industrial fluid systems, where sealing integrity, flow control, and safe shut-off are essential requirements. They play a critical role in oil and gas, water distribution, and process industries, where even minor sealing failures can have significant safety and operational consequences. Mottahedin et al. (2024) demonstrate that isolation valves are not passive assets but integral components of maintenance planning and network reliability analysis.

Valves degrade over time through interacting mechanisms, including seat wear, sealing surface deterioration, pressure imbalances, and erosion-corrosion damage. Feng et al. (2025) emphasize that such degradation is progressive and accumulative, making early detection particularly important. Crucially, as Bhilare et al. (2025) show, pressure and flow signals recorded by monitoring systems are not abstract data streams but direct reflections of the valve's fluid mechanics, structural response, and sealing condition.

This dissertation analyses the condition monitoring of gate valves using low-frequency time-series data and high-frequency sensor data from real qualification tests conducted by TechnipFMC. The study focuses on cycle-based KPI analysis, degradation assessment, and high-frequency signal characterization using RMS, kurtosis, power spectral density, and spectrogram methods.

## 1.2 Problem Statement

Conventional valve evaluation relies primarily on threshold-based acceptance criteria, yielding a binary pass or fail outcome. While useful, this approach does not indicate how the valve performed during the test, whether behaviour was repeatable, or whether any progressive degradation was occurring. Han et al. (2020) argue that leakage assessment should draw on quantitative diagnostic analysis rather than outcome-based interpretation alone. The fundamental problem addressed by this dissertation is the absence of a unified analytical framework that connects low-frequency operational data with high-frequency sensor behaviour to provide a more detailed and physically meaningful assessment of valve health.



### **1.3 Research Aim**

This dissertation aims to develop and apply a data-driven framework for the condition monitoring and performance analysis of industrial gate valves, using low-frequency time-series data and high-frequency sensor data from qualification testing, in order to identify indicators of valve behavior, repeatability, degradation, and potential fault signatures.

### **1.4 Research Objectives**

To achieve this aim, five objectives are pursued:

1. To analyze gate valve actuation cycles using low-frequency time-series data across multiple pressure classes.
2. To extract key performance indicators reflecting sealing quality and operational condition.
3. To evaluate cycle repeatability and identify potential degradation trends across 500 actuation cycles.
4. To analyze high-frequency sensor data using RMS, kurtosis, PSD, and spectrogram methods.
5. To identify characteristic operational signatures and detect anomalies associated with abnormal valve behaviour or emerging faults.

### **1.5 Research Questions**

6. How can low-frequency time-series data characterize gate valve actuation cycles and operational performance?
7. Which KPIs are most useful for evaluating sealing behaviour, pressure stability and leakage-related change?
8. To what extent does valve performance vary or degrade across repeated actuation cycles?
9. How can high-frequency features such as RMS, kurtosis, PSD and STFT contribute to condition monitoring?
10. Can low-frequency and high-frequency data together distinguish normal from anomalous or degrading behaviour?

## **1.6 Rationale and Significance**

In industrial terms, gate valves are deployed in systems where reliability, containment, and safe isolation are critical. Mottahedin et al. (2024) clarify that valve decisions affect maintenance planning and network performance, while Feng et al. (2025) emphasise progressive degradation under service conditions. More informative monitoring strategies are therefore consistent with earlier intervention and more defensible maintenance decisions.

In academic terms, the research sits at the boundary of mechanical engineering and data analytics. Qi, Zhang, and Spencer (2023) note a general increase in data-driven condition monitoring, but gate valve studies remain sparse relative to work on pumps, bearings, and rotating machinery. This dissertation contributes to that emerging field.

## **1.7 Scope of the Study**

This dissertation analyses experimental qualification test data for gate valves at 5k, 10k, and 15k psi pressure classes, provided by TechnipFMC. The scope covers low-frequency hydraulic cycle data from 500-cycle leakage tests on each valve class, and high-frequency sensor data (accelerometer at 400 Hz, magnetometer at 100 Hz) from a 92-cycle HF vibration test. All analyses are implemented in MATLAB. The study does not attempt full-scale field deployment, real-time implementation, or a complete machine learning classification system.

## **1.8 Dissertation Structure**

Chapter 1 introduces the research background, problem, aim, and objectives. Chapter 2 reviews literature on gate valve operation, failure mechanisms, condition monitoring, and signal analysis methods. Chapter 3 describes the methodology. Chapter 4 presents and discusses results. Chapter 5 draws conclusions and proposes future work.

## **1.9 Chapter Summary**

This chapter has established the engineering context and problem motivating the study, stated the aim, objectives, and research questions, and outlined the rationale, scope and structure. Chapter 2 reviews the supporting literature.

## **Chapter 2: Literature Review**

### **2.1 Introduction**

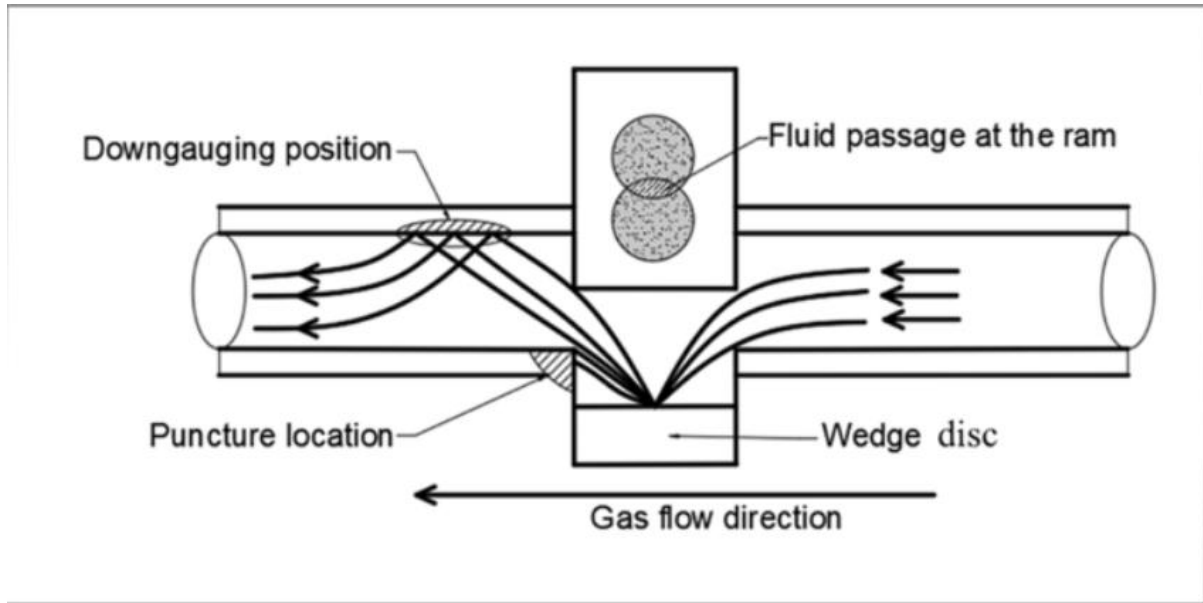
This chapter reviews literature relevant to the condition monitoring and performance analysis of gate valves using time-series and high-frequency sensor data. It begins with valve function and failure, transitions to condition monitoring and signal analysis, and concludes by identifying the research gap and the conceptual framework underpinning the study. While substantial literature exists on valve behaviour and on broader data-driven monitoring, studies combining low-frequency operational measurements with high-frequency sensor analysis specifically for gate valves remain limited, justifying the current study (Mottahedin et al., 2024; Qi, Zhang, and Spencer, 2023).

### **2.2 Gate Valves in Industrial Systems**

Gate valves function as isolation devices in industrial fluid systems, providing full unobstructed bore flow when open and positive shut-off when closed. They are critical in oil and gas pipelines, process plants, and water distribution networks. Mottahedin et al. (2024) demonstrate that isolation valves are essential elements of maintenance decision-making, not merely passive assets. Bhilare et al. (2025) show through experimental and numerical investigation that flow through a gate valve is shaped by geometry and operating conditions in ways that influence the pressure and flow signals recorded by a monitoring system, grounding sensor measurements in physical valve engineering.

### **2.3 Failure Mechanisms and Leakage Behaviour in Gate Valves**

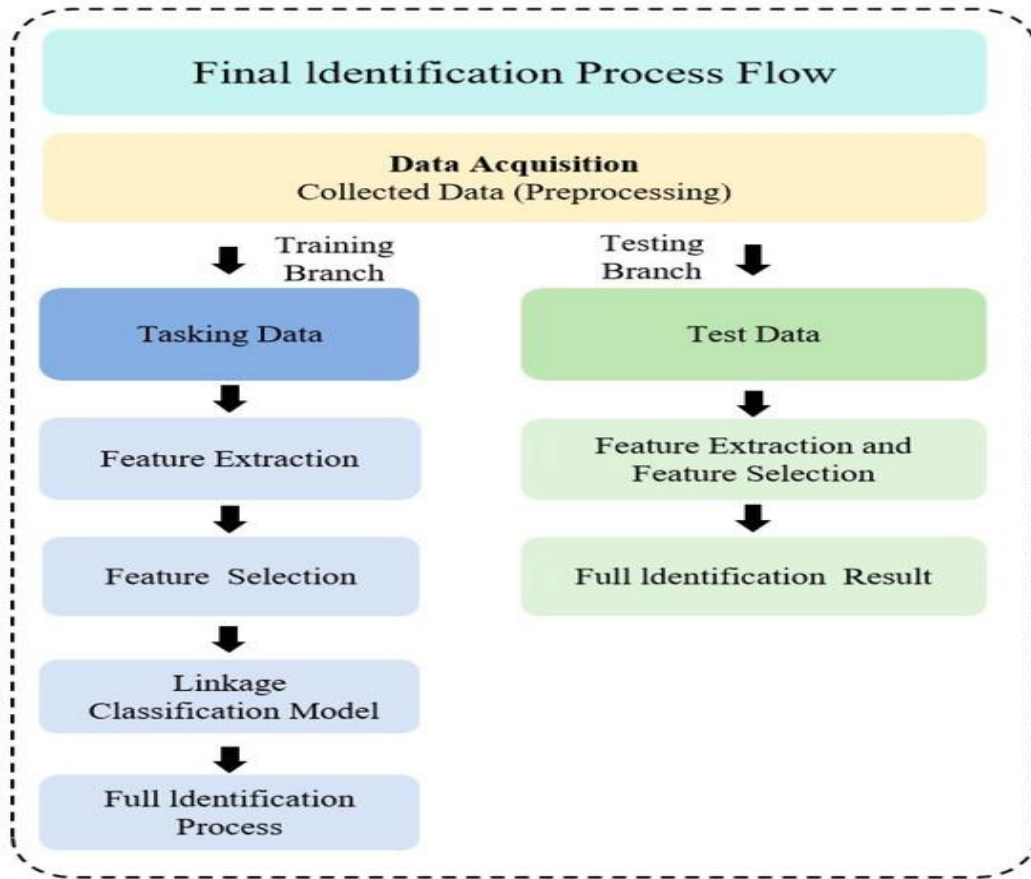
Gate valve failure typically arises from interacting mechanisms: wear, erosion, corrosion, seat damage, sealing surface deterioration, and repeated mechanical loading. Feng et al. (2025) find that erosion-corrosion coupling produces progressive structural degradation, confirming that valve degradation is accumulative. Leakage is the most significant practical manifestation. Han et al. (2020) argue that leakage is a quantitative diagnostic problem rather than a binary threshold, while Wang et al. (2025) demonstrate that internal leakage can be studied using ultrasonic sensing in both liquid and gas environments. Zagretdinov et al. (2024) and Ye, Xu and Wu (2021) confirm that acoustic and acoustic-emission features support leakage detection, supporting the multi-domain approach taken in this dissertation.



*[Figure 2.1: Gate valve degradation mechanisms and diagnostic indicators (adapted from Feng et al., 2025; Han et al., 2020; Zagretdinov et al., 2024; Ye, Xu and Wu, 2021; Wang et al., 2025).]*

## 2.4 Condition Monitoring in Mechanical and Fluid Systems

Condition monitoring measures system parameters to assess equipment health during operation or testing. Ul Hassan, Panduru, and Walsh (2025) identify it as a foundational component of Industry 4.0, and Qi, Zhang, and Spencer (2023) demonstrate that data-driven monitoring encompasses statistical, mathematical, and learning-based approaches. Most established literature, however, addresses rotating machinery rather than valves. Valve faults can manifest differently: as pressure decay, cycle-to-cycle inconsistency, or brief transient events rather than continuous vibration signatures. This distinction makes valve-oriented condition monitoring a distinct and emerging research area.



[Figure 2.2: Condition monitoring framework for gate valve performance assessment (adapted from Chen et al., 2022; Qi, Zhang and Spencer, 2023; Ul Hassan, Panduru and Walsh, 2025).]

## 2.5 Time-Series Analysis for Valve Performance Assessment

Time-series analysis is central to gate valve monitoring. Beyond visualization, it enables cycle isolation, stability comparison, and detection of systematic behavioural change. Madhavan, Shahgaldi, and Li (2025) demonstrate that valve pressure responses can be modelled as structured time patterns across long lifecycles. A cycle-based approach is particularly well-suited to this dissertation: test signals exhibit distinct phases (pressurization, hold, depressurization, exhaust) whose individual examination yields more diagnostic information than treating the signal as a single continuous trace. Han et al. (2020) confirm that leakage diagnosis is enhanced when signal behaviour is interpreted against system characteristics rather than endpoint results alone.

## 2.6 Key Performance Indicators for Valve Condition Evaluation

KPIs distil large sensor datasets into targeted metrics quantifying specific aspects of valve behaviour. Ferrarese and Malavasi (2022) demonstrate that clearly defined performance

parameters enhance condition assessment when operating conditions vary. KPI selection requires care: peak hydraulic pressure confirms actuation was achieved but reveals little about sealing; pressure decay rate is more informative for leakage but requires a consistently defined observation window; minimum flow rate may indicate leakage but may also reflect sensor sensitivity or operational variations. KPIs are most valuable when they simplify data without severing the connection to valve physics.

## **2.7 High-Frequency Analysis in Fault Detection**

Low-frequency operational data does not always capture short-duration or impulsive faults. Wang et al. (2025) confirm that internal leakage can be studied using ultrasonic signals, treating leakage as a signal detection problem. Zagretdinov et al. (2024) and Ye, Xu and Wu (2021) show that acoustic and acoustic-emission features can detect internal leakage. Together, these studies establish high-frequency sensing as an important dimension of any comprehensive valve condition monitoring system.

## **2.8 Time-Domain Features: RMS and Kurtosis**

RMS quantifies signal power or average magnitude. Increased RMS in vibration monitoring typically indicates more vigorous dynamic activity. Kurtosis quantifies impulsiveness: the degree to which a signal contains extreme spikes relative to its baseline distribution. For a Gaussian signal, kurtosis equals 3; values substantially above 3 indicate impact-like events or transients. Venkata and Rao (2019) demonstrate that vibration-based time-domain features classify healthy and faulty valve conditions, supporting their diagnostic utility. The literature also cautions that high RMS or kurtosis may arise from legitimate operational events or test artefacts; interpretation alongside physical timing and complementary channels is essential.

## **2.9 Frequency-Domain and Time-Frequency Analysis: PSD and Spectrogram**

Power spectral density (PSD) reveals how signal energy is distributed across frequencies, identifying dominant components and spectral changes associated with mechanical events. Venkata and Rao (2019) apply frequency-domain analysis to control valve fault identification, and Mercorelli (2024) confirms the continuing relevance of spectral techniques in diagnostic systems. Gate valve signals are non-stationary; their frequency content changes during actuation events. STFT spectrograms address this by revealing when specific frequency components are active, providing temporal resolution that a single PSD cannot offer.

Zagretdinov et al. (2024) and Ye, Xu and Wu (2021) confirm that temporal structure in high-frequency signals is relevant to leakage detection.

## **2.10 Data Pre-Processing and Signal Segmentation**

Raw sensor data requires substantial preparation before analysis. In engineering datasets, pre-processing is where the real analytical work begins. Signals require noise handling, variable naming must be standardized, time axes reconstructed, and event windows isolated before feature extraction. Qi, Zhang, and Spencer (2023) recognize data handling as integral to any monitoring framework, echoed by Ul Hassan, Panduru, and Walsh (2025). Signal segmentation is particularly critical: valve data contains repeated cycles, but until boundaries are precisely identified, the data has limited analytical value. Pre-processing and segmentation are therefore core analytical activities, not peripheral ones.

## **2.11 Research Gap**

The review confirms the importance of valve reliability, leakage assessment, sensor-based monitoring, and signal analysis. However, the intersection of these strands is limited. Gate valve studies tend to focus on flow behaviour, structural degradation, or single-mode leakage detection, while data-driven monitoring reviews draw predominantly on rotating machinery examples. Minimal integrated research combines low-frequency hydraulic KPI analysis with high-frequency sensor characterization in a unified gate valve monitoring framework. This dissertation aims to address that gap by connecting operational and dynamic information in a physically grounded way.

## **2.12 Conceptual Framework for the Present Study**

The conceptual framework rests on three premises. First, valve condition is manifested in quantifiable signal behaviour. Second, different signal types reveal different aspects of the condition: low-frequency variables reveal cycle structure, pressure stability, and leakage drift, while high-frequency signals reveal transient and impulsive behaviour. Third, extracted features gain significance when interpreted in relation to the physical test process and valve engineering context, not as isolated statistics. The dissertation, therefore, adopts a combined engineering and data perspective, treating the valve as a physical system whose health can be inferred from multi-scale signal behaviour.

### **2.13 Chapter Summary**

This chapter reviewed literature on gate valve condition monitoring and performance analysis. It has been established that valve behaviour is governed by complex flow, sealing, and degradation interactions; that leakage is best understood through multiple complementary indicators; and that both low-frequency operational measurements and high-frequency signals carry diagnostic value. A research gap was identified in the absence of integrated gate valve monitoring frameworks combining cycle-based analysis with high-frequency feature extraction. Chapter 3 describes the methodology developed to address this gap.



## **Chapter 3: Methodology**

### **3.1 Introduction**

This chapter describes the data sources, pre-processing procedures, cycle segmentation methods, KPI extraction approaches, and signal analysis techniques used in this dissertation. The methodology is structured around two datasets reflecting two experimental programs: cyclic pressure (leakage) tests and high-frequency (HF) vibration tests. For each dataset, pre-processing challenges are described before analytical methods are introduced, reflecting the reality that robust pre-processing is a prerequisite for meaningful analysis (Qi, Zhang and Spencer, 2023). All analyses were implemented in MATLAB without proprietary toolboxes.

### **3.2 Data Sources and Test Program**

#### **3.2.1 Industry Partner and Test Context**

Experimental data were provided by TechnipFMC as part of an MSc industrial collaboration. The data originates from gate valve qualification trials conducted in accordance with API 6A / ISO 10423 standards, which specify endurance testing requirements for high-pressure wellhead and surface equipment. Two test types were performed: cyclic pressure (leakage) tests, which subject each valve to repeated open-close cycles at rated pressure to evaluate seat leakage performance, and HF vibration tests, which apply controlled shaker-induced vibration during the pressure hold phase to simulate the operational pipeline vibration environment.

#### **3.2.2 Cyclic Pressure Test Datasets**

Three valve pressure classes were tested: 5,000 psi (5k), 10,000 psi (10k) and 15,000 psi (15k), each undergoing 500 open-close cycles. Data were recorded at approximately 10 Hz per cycle across ten CSV files per valve (50 cycles per file). The four principal channels were: hydraulic actuator pressure (psig), differential pressure across the seat (psid), Side A line pressure (psig), and flowmeter reading (gpm). The 10k dataset included additional channels (bore pressure, cavity pressure, temperature) not present in the 5k dataset.

#### **3.2.3 High-Frequency Vibration Test Dataset**

The HF test dataset covers 92 valve actuation cycles recorded over three consecutive days (24-26 November 2025). The primary file (A03 - Acc and Mag Data with cycle markers.txt, 2.8 GB) contains eight channels: accelerometer axes AccX, AccY, AccZ at 400 Hz, and

magnetometer axes MagX, MagY, MagZ at 100 Hz (every fourth row). An EVENTS column, added by the TechnipFMC engineer, contains one of four markers per row where relevant: open cycle start, open cycle end, close cycle start, and close cycle end. A separate pressure file records actuator, Side A and Side B pressures at approximately 1 Hz over 4.6 hours of testing.

### **3.3 Pre-Processing: Cyclic Leakage Tests**

#### **3.3.1 File Combination and Time-Axis Reconstruction**

The ten CSV files per valve contained inconsistent column naming and non-resetting timestamps across file boundaries. All files were loaded and concatenated in sequence, with timestamps reconstructed to produce a continuous time axis from cycle 1 to cycle 500. Inconsistent column headers were standardized before concatenation.

#### **3.3.2 Cycle Detection**

No explicit cycle labels were present in the raw data. Cycles were detected algorithmically using threshold crossings on the hydraulic pressure signal, defining each cycle as the period from pressurization onset to depressurization completion. Edge cases at file boundaries and isolated pressure spikes were identified through visual inspection and handled manually. Total detected cycle counts were verified against the expected 500 per valve class.

#### **3.3.3 Outlier and Noise Handling**

Cycles exhibiting isolated pressure dips or flow anomalies were cross-referenced across multiple sensor channels to distinguish genuine valve degradation from external test-rig disturbances. Potentially anomalous cycles were retained in the dataset but annotated for further investigation.

### **3.4 Pre-Processing: High-Frequency Vibration Test**

#### **3.4.1 Large File Extraction**

The 2.8 GB accelerometer file was too large to load directly into memory. Individual cycle windows were extracted using PowerShell command-line tools, selecting specific row ranges calculated from the EVENTS timestamps with 15-second padding on each side. Row positions were calculated as:  $\text{Row number} = (\text{EVENTS timestamp} - \text{Accelerometer recording start}) \times 400 \text{ Hz}$ , where the accelerometer recording start was confirmed as 11:56:50 on 24 November 2025.

### 3.4.2 Gravity and DC Bias Removal

AccX exhibited a constant offset of approximately  $9.81 \text{ m/s}^2$  due to gravitational acceleration in the sensor mounting orientation. AccY and AccZ showed smaller DC offsets from imperfect levelling. All three axes were corrected by subtracting the channel mean over the full cycle window, isolating the dynamic vibration signal from static sensor bias.

### 3.4.3 Pressure File Time Alignment

Pressure file timestamps were stored in MM:SS format only, introducing hour-boundary ambiguity. A rollover detection algorithm reconstructed the full-time axis, incrementing the hour count when the minute value decreased from above 50 to below 10. The pressure file was confirmed to start at 13:58:48 on 24 November 2025, yielding an alignment offset of 272 seconds relative to the accelerometer recording start. Pressure times in accelerometer coordinates were calculated as:  $T_{\text{pressure\_acc}} = (T_{\text{pressure\_raw}} - T_{\text{pressure\_raw}(1)}) + 272 \text{ seconds}$ .

### 3.4.4 Valve Position Signal (MagZ)

The MagZ channel was used as a valve position indicator, confirmed by the TechnipFMC engineer: MagZ maximum corresponds to fully open and MagZ minimum to fully closed. MagZ was recorded at 100 Hz (every fourth row) and linearly interpolated to the 400 Hz accelerometer grid for analysis. The normalized position signal was computed as:  $\text{MagZ\_norm} = (\text{MagZ} - \min(\text{MagZ})) / (\max(\text{MagZ}) - \min(\text{MagZ}))$ , yielding 0 (fully closed) to 1 (fully open).

## 3.5 Cycle Segmentation

### 3.5.1 Cyclic Leakage Tests

Cycles were segmented using hydraulic pressure threshold crossings as described in Section 3.3.2. The four test phases within each cycle (depressurization, pressurization, hold, exhaust) were identified from characteristic pressure patterns and used to define KPI extraction windows.

### 3.5.2 High-Frequency Vibration Test

Cycle segmentation used the EVENTS markers provided by the TechnipFMC engineer. Each complete cycle was defined from open cycle start to close cycle end, with opening and closing phases separately delineated by the intermediate markers. This approach provides precise, physically verified cycle boundaries enabling separate analysis of opening (gate travelling closed to open) and closing (gate travelling open to closed) phases. Stroke durations for all 92

cycles were calculated directly from the EVENTS timestamps. Total span covers 47.2 hours with overnight interruptions visible between cycles 20-21 and 70-71.

### **3.6 Key Performance Indicator Extraction**

#### **3.6.1 Cyclic Leakage Test KPIs**

Four primary KPIs were extracted for each of the 500 cycles per valve class:

- Peak hydraulic pressure (psig): maximum actuator pressure during pressurization, confirming whether the target test pressure was achieved.
- Peak differential pressure (psid): maximum pressure difference across the seat, reflecting the sealing load.
- Average flow rate at high pressure (gpm): mean flowmeter reading during the hold period, proxy for seat leakage rate.
- Minimum flow rate at high pressure (gpm): minimum flowmeter reading during hold, lower bound on leakage.

For each KPI, a 25-cycle rolling mean and plus or minus one standard deviation envelope were calculated to distinguish systematic trends from cycle-to-cycle noise. Linear trend lines were fitted across the 500-cycle window to quantify rate of change per cycle.

#### **3.6.2 High-Frequency Vibration Test KPIs**

Stroke duration KPIs were extracted from EVENTS timestamps for all 92 cycles:

- Opening stroke duration (seconds): open cycle start to open cycle end.
- Closing stroke duration (seconds): close cycle start to close cycle end.
- Total cycle duration (seconds): open cycle start to close cycle end.

Descriptive statistics (mean, standard deviation, minimum, maximum) were computed per test day to identify systematic progression across the three-day test period.

### **3.7 Signal Analysis Methods: High-Frequency Vibration**

#### **3.7.1 RMS and Kurtosis**

RMS and Pearson kurtosis were calculated for each accelerometer axis over the opening and closing phases separately. RMS was computed as the square root of the mean of squared values. Kurtosis was computed as the fourth standardized central moment:  $\text{mean}((x - \text{mean}(x))^4) / \text{std}(x)^4$ . For a Gaussian signal, kurtosis equals 3; values substantially above 3 indicate

impulsive behaviour. Rolling versions of both metrics were computed using a one-second (400-sample) sliding window to characterize within-cycle evolution.

### **3.7.2 Power Spectral Density (Welch Method)**

PSD was estimated using Welch's method with segment length  $L = 2048$  samples (frequency resolution approximately 0.195 Hz), 50% overlap (1024 samples), and a Hann window. The average one-sided PSD was plotted on a logarithmic scale (dB re 1  $(\text{m/s}^2)^2/\text{Hz}$ ) against frequency from 0 to 100 Hz.

### **3.7.3 Short-Time Fourier Transform (Spectrogram)**

STFT analysis was applied to resolve how vibration frequency content evolved through each cycle. Parameters: segment length 512 samples (1.28 seconds), 480-sample overlap (93.75%), Hann window, yielding time resolution approximately 0.08 seconds and frequency resolution approximately 0.78 Hz. Results displayed as dB-scaled spectrograms from 0 to 100 Hz, with time axes aligned to the common cycle time reference shared by MagZ, accelerometer, and pressure panels.

## **3.8 Analytical Framework Summary**

The analytical approach integrates multiple complementary methods. For cyclic leakage tests, KPI-based trend analysis detects cycle-to-cycle degradation and anomalies. For the HF vibration test, MagZ-based segmentation provides precise phase boundaries, stroke duration KPIs track performance across all 92 cycles, and time-domain (RMS, kurtosis), frequency-domain (PSD) and time-frequency (STFT) analysis characterize vibration within individual cycles. Both analytical streams are grounded in the physical engineering context of the valve test and interpreted in relation to known test conditions and engineer-confirmed observations.

## **3.9 Chapter Summary**

This chapter describes data sources, pre-processing procedures, segmentation approaches, KPI extraction methods, and signal analysis techniques. The principal pre-processing challenges — large file size, timestamp truncation, gravity removal, and absence of explicit cycle labels in the original data — were each systematically addressed. Chapter 4 presents and discusses the results obtained by applying this methodology to both datasets.

## **Chapter 4: Results and Discussion**

### **4.1 Introduction**

This chapter presents and discusses the results of the analysis. Section 4.2 covers the cyclic pressure (leakage) test analysis for the three valve classes. Section 4.3 covers the high-frequency vibration test analysis. Results are presented progressively from overview-level observations to detailed anomaly investigation, with findings discussed in relation to physical valve behaviour and the literature reviewed in Chapter 2.

### **4.2 Cyclic Pressure (Leakage) Test Analysis**

#### **4.2.1 5k Valve — KPI Trends and Test Failure**

The 5,000 psi class valve was the only valve to fail its qualification test. KPI trend analysis across 500 cycles revealed a clear and progressive deterioration signature beginning around cycle 370-400.

Peak hydraulic pressure remained stable at approximately 1,200 psig from cycles 1 through 400, before a sudden collapse beginning at cycle 408. By cycles 425-430, hydraulic pressure had dropped below 700 psig and continued declining, indicating that the system could no longer maintain target test pressure. A linear decline of approximately 9.74 psig per cycle was confirmed across the analysis window.

The most diagnostic signal was differential pressure. This KPI remained stable at approximately 80 psid throughout cycles 1-428, then collapsed to near zero at cycles 429-430. A differential pressure of zero indicates complete loss of pressure difference across the valve seats, the operational definition of seat leakage failure. This is consistent with Han et al. (2020), who characterize leakage as a progressive quantitative problem leading ultimately to loss of seating integrity.

Average and minimum flow rate KPIs showed small positive upward trends (approximately +0.005 and +0.011 gpm per cycle, respectively) prior to the main collapse, suggesting a slow leakage increase before the more dramatic pressure failure. Early cycles (1-50) also showed elevated flow rates that stabilized over 10-15 cycles, consistent with a component bedding-in period at the start of testing.



*[Figure 4.1—5k valve KPI trends, cycles 370-430, showing pressure collapse and differential pressure failure at cycles 429-430.]*

#### 4.2.2 10k Valve — KPI Trends and Test Pass

The 10,000-psi class valve passed its qualification test. KPI analysis revealed a deliberate test condition step-change at cycle 55 that initially appeared anomalous but was interpretable as a controlled test pressure escalation.

Peak hydraulic pressure was stable at approximately 710 psig from cycles 44 through 53, then stepped sharply to approximately 1,200 psig at cycle 55. This step-change reflects the test engineer increasing the hydraulic supply pressure to advance to a higher severity test phase. Differential pressure showed a marginal downward trend (-0.25 psid per cycle) but remained within specification throughout.

The minimum flow rate KPI was most diagnostic. From cycles 44-53, minimum flow was near zero gpm, indicating effective sealing at the lower test pressure. Following the step-change at cycle 55, minimum flow increased to approximately 22-23 gpm — a response to the greater pressure differential driving leakage. This level remained within the specified acceptance limits for the 10k class, explaining the test pass. The near-zero minimum flow around cycle 50, therefore indicates the valve sealing well, not a valve fault.



[Figure 4.2— 10k valve KPI trends cycles 44-56, showing deliberate test condition step-up at cycle 55.]

#### 4.2.3 15k Valve — KPI Trends and Test Pass

The 15,000 psi class valve completed 500 cycles and passed qualification. KPI trends were stable throughout with no significant degradation in any primary indicator. The additional sensor channels available (bore pressure, cavity pressure, temperature) provided richer diagnostic information than the lower-class datasets. The 15k valve dataset provides a useful baseline of stable valve behavior against which the anomalous signatures in the 5k and 10k datasets can be contextualized.

#### 4.2.4 Cross-Class Comparison

Comparing KPI behavior across the three valve classes highlights a clear contrast: the 5k valve exhibited an irreversible pressure collapse from cycle 408; the 10k valve showed a deliberate test escalation at cycle 55 but maintained acceptable leakage; and the 15k valve remained stable throughout. These results demonstrate that the KPI framework successfully discriminates between normal, anomalous, and degrading behavior within a qualification test context, extending the binary pass/fail interpretation to provide mechanistic insight into when and how degradation occurs.

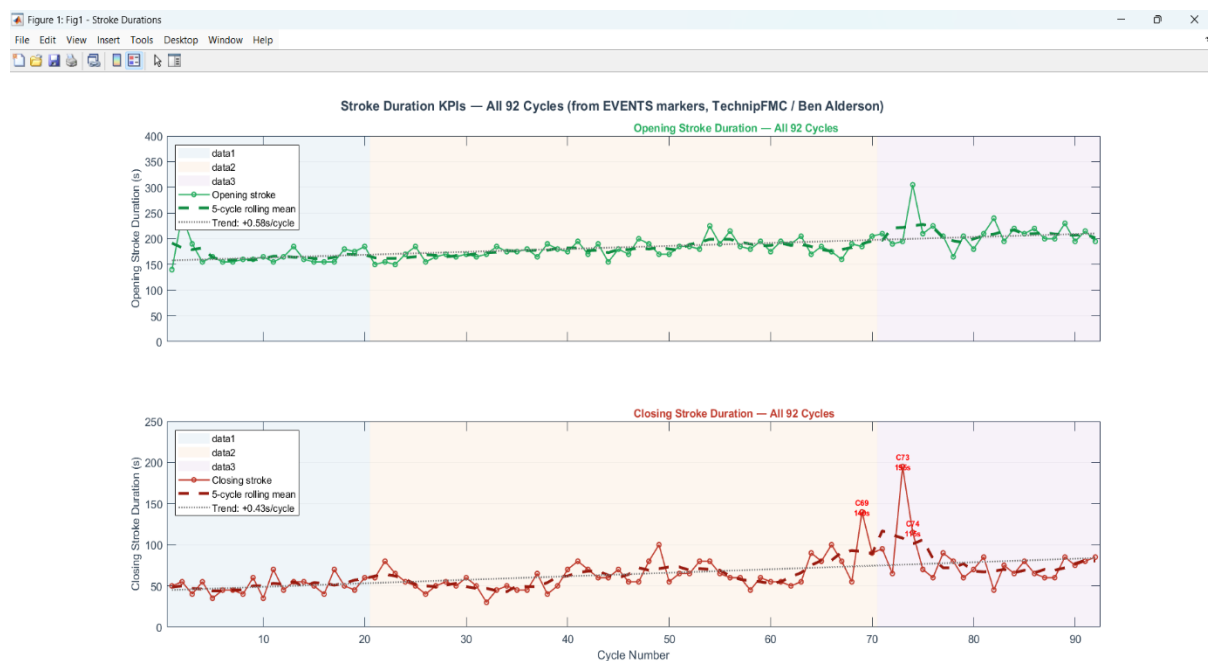


## 4.3 High-Frequency Vibration Test Analysis

### 4.3.1 Stroke Duration KPIs — All 92 Cycles

Stroke duration KPIs extracted from the EVENTS markers for all 92 cycles reveal a consistent and statistically significant progressive increase in both opening and closing stroke durations across the three test days.

Mean opening stroke duration increased from 168 seconds on Day 1 (cycles 1-20) to 179 seconds on Day 2 (cycles 21-70) and 210 seconds on Day 3 (cycles 71-92), a 25% increase from Day 1 to Day 3. Mean closing stroke duration increased from 50 seconds on Day 1 to 63 seconds on Day 2 and 80 seconds on Day 3, a 60% increase. Linear trend fits confirmed systematic rather than random progression across both metrics.



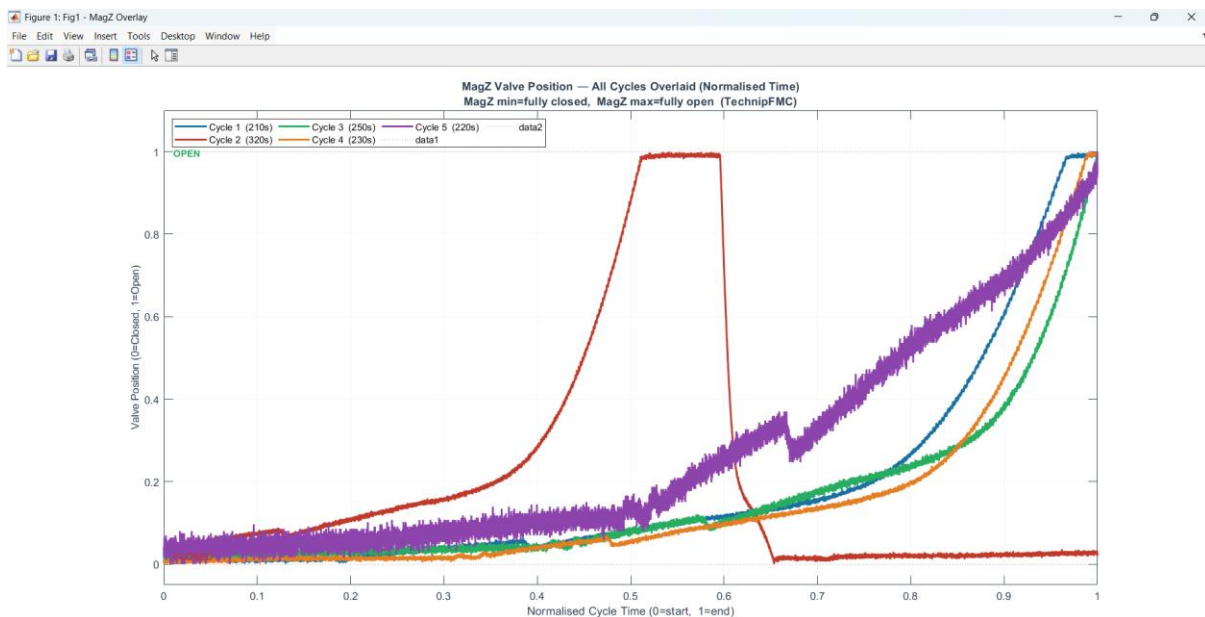
[Figure 4.3 — stroke duration KPIs across all 92 cycles with day shading and rolling means, showing progressive increase from Day 1 to Day 3.]

The TechnipFMC engineer identified several potential contributing factors: depletion of compressed air supply pressure from the MCP cylinder palette over the three-day test (which would reduce HPU flow rate and slow the opening stroke); thermal effects in the hydraulic fluid (expected to stabilize within a few cycles each day rather than persist across days); and friction or contamination changes. The engineer noted that mechanical wear is unlikely to be the primary cause since opening times returned to normal after anomalous cycles, suggesting a recoverable rather than permanent change.

The most anomalous individual cycles were Cycle 73 (closing 195 seconds, Day 3), Cycle 69 (closing 140 seconds, Day 2), Cycle 74 (opening 305 seconds — the longest of any cycle — with closing 115 seconds, Day 3), and Cycles 49 and 66 (closing 100 seconds each, Day 2). Extended closing times in these cycles were attributed by the engineer to the dump valve not being fully opened by the test technician on those occasions.

### 4.3.2 MagZ Valve Position Analysis

The normalized MagZ signal provided a continuous record of valve gate position throughout each actuation cycle, ranging from 0 (fully closed) to 1 (fully open). Analysis of cycles 1-5 confirmed the asymmetric actuation mechanism. Opening strokes lasted approximately 160-250 seconds, consistent with the HPU specification of 0.72 liters per minute maximum flow rate, gradually filling the actuator cylinder. Closing strokes under normal conditions lasted 2-8 seconds, consistent with spring-return actuation.



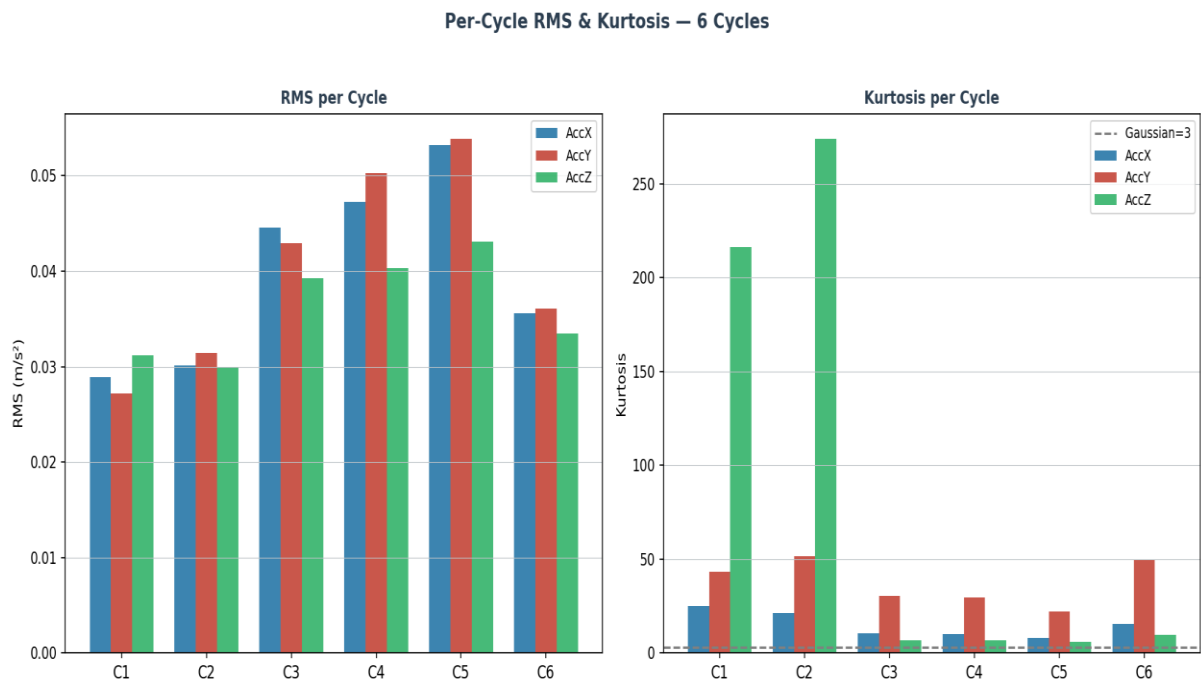
*[Figure 4.4 — MagZ normalized valve position overlay for cycles 1-5, showing consistent opening profile and fast closing (except Cycle 2).]*

Cycle 2 was identified as anomalous with a closing stroke duration of 158 seconds, compared to 2-8 seconds for all other cycles in the group. The MagZ signal confirmed genuine slow gate travel during closure rather than a sensor artefact, with multiple MagZ oscillations visible during the extended closing period. The engineer attributed this to the dump valve not being fully opened on that occasion.

### 4.3.3 Vibration Analysis — RMS and Kurtosis

RMS and kurtosis analysis across cycles 1-5 revealed a clear and physically interpretable transition in vibration character between early and later cycles. AccZ kurtosis values were 216 for Cycle 1 and 274 for Cycle 2, compared to 6-10 for Cycles 3-5 — approximately 70-90 times above the Gaussian baseline of 3, indicating highly impulsive vibration in the early cycles. Conversely, AccZ RMS was lower in cycles 1-2 (approximately 0.030-0.031 m/s<sup>2</sup>) than in cycles 3-5 (0.039-0.043 m/s<sup>2</sup>), indicating that later cycles had more total vibration energy despite being less impulsive.

This pattern — decreasing kurtosis and increasing RMS across cycles — is consistent with a component bedding-in process, confirmed by the TechnipFMC engineer. Initial actuations produce sharp mechanical impacts as seating surfaces make first contact (high kurtosis, moderate RMS). As cycles progress, surfaces settle into repeatable contact and vibration becomes sustained and broadband (lower kurtosis, higher RMS). This interpretation is physically plausible and consistent with vibration-based condition monitoring literature (Venkata and Rao, 2019).

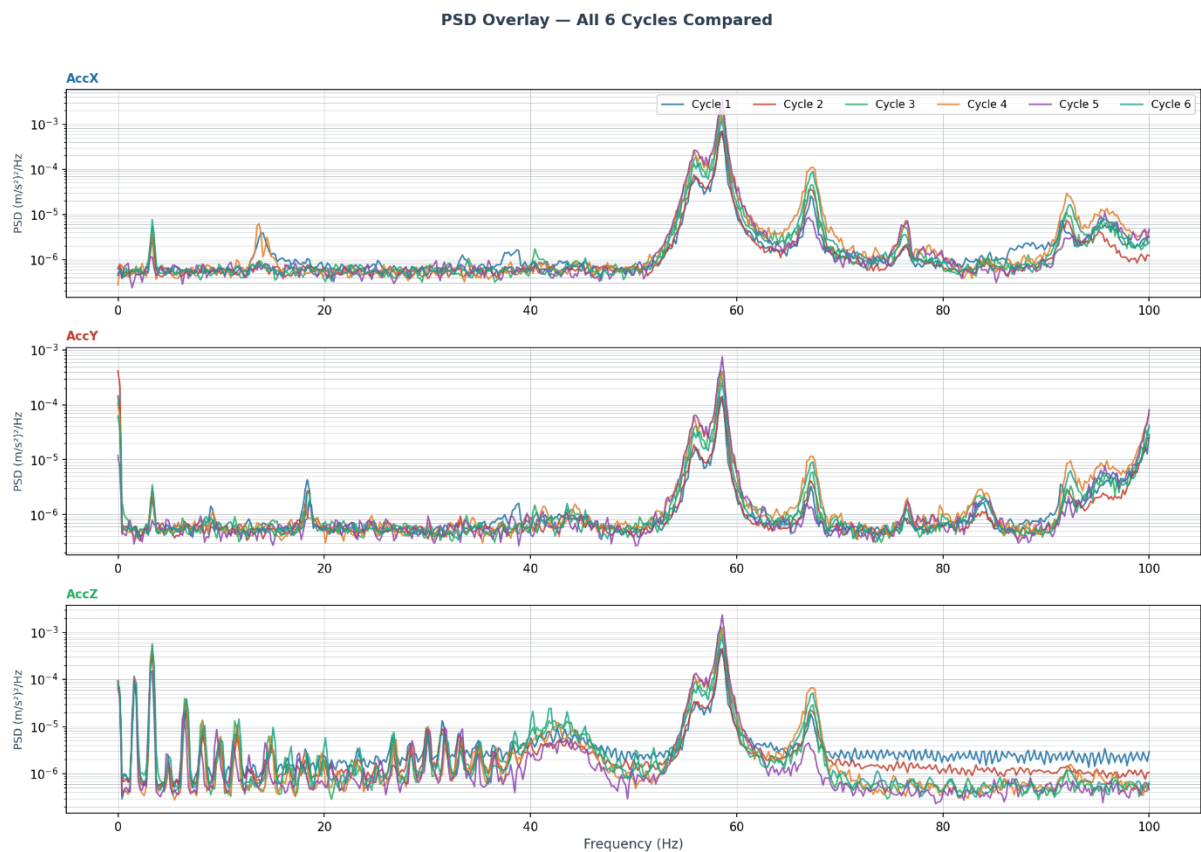


*[Figure 4.5 — per-cycle RMS and kurtosis bar chart for cycles 1-5 across AccX, AccY and AccZ, showing decreasing kurtosis and increasing RMS trend.]*

#### 4.3.4 Power Spectral Density Analysis

PSD overlay analysis for cycles 1-5 confirmed a dominant spectral peak at approximately 58 Hz across all cycles and all three accelerometer axes. This corresponds to the shaker excitation frequency, confirming consistent test conditions were maintained across cycles. Secondary peaks at approximately 75 Hz and 95 Hz are attributed to structural resonances of the valve body or test rig.

The key inter-cycle difference is most visible in AccZ at low frequencies (0-20 Hz). Cycles 1 and 2 exhibit noticeably higher spectral density in this region compared to cycles 3-5. Elevated low-frequency content corresponds to the impulsive mechanical shocks identified by kurtosis analysis: sharp time-domain spikes produce broadband low-frequency energy in the frequency domain. By cycles 3-5, low-frequency content has reduced substantially and spectral profiles converge, consistent with post-bedding-in steady-state behaviour.

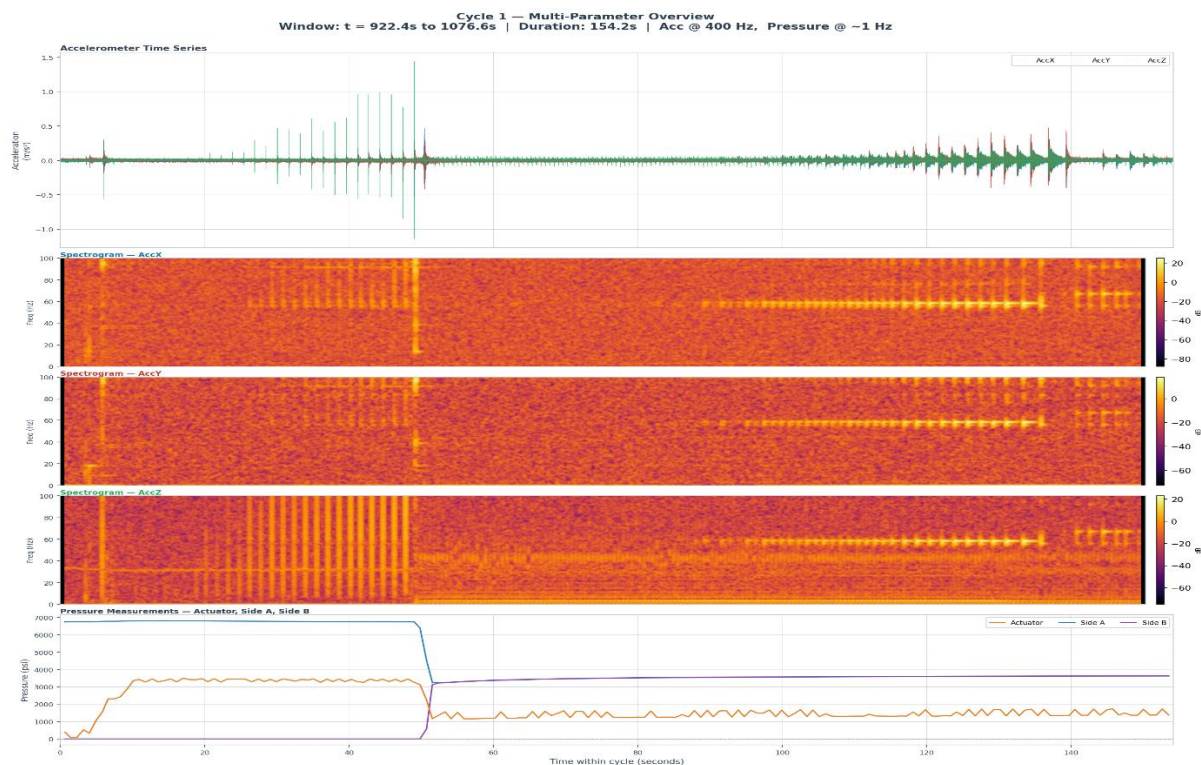


*[Figure 4.6 — PSD overlay for cycles 1-5, AccX/AccY/AccZ, showing dominant 58 Hz shaker peak and elevated low-frequency energy in cycles 1-2.]*

#### 4.3.5 STFT Spectrogram and Multi-Parameter Analysis

STFT spectrogram analysis provided the clearest visualization of temporal evolution in vibration behavior. Multi-parameter plots combining MagZ position, AccX, AccY, AccZ spectrograms, and actuator pressure on a shared time axis enabled direct correlation between valve position and vibration character. This approach directly addressed the request from Professor Wallace for a consolidated per-cycle view of all sensor channels.

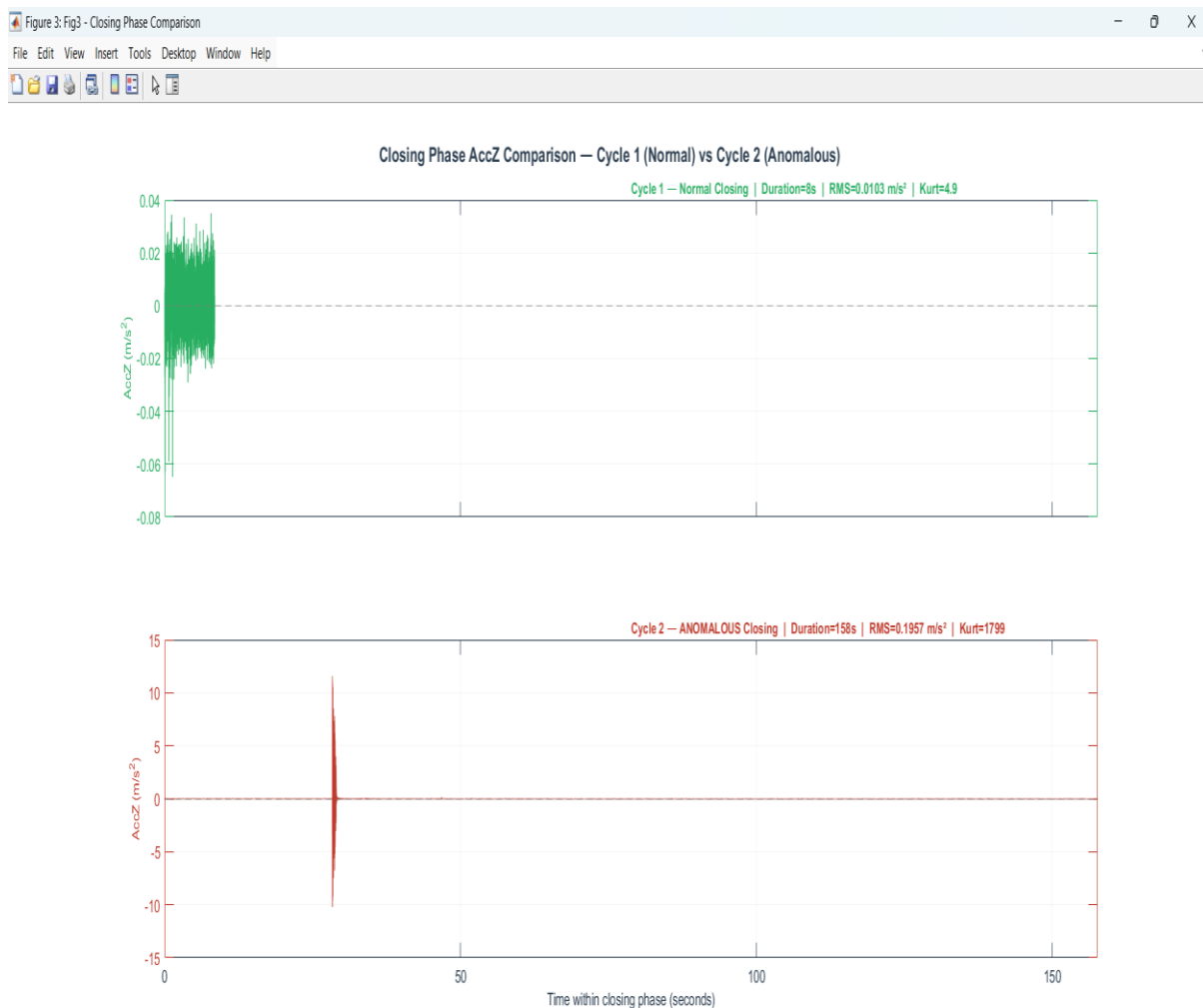
For Cycle 1, the spectrograms show a clear transition at approximately 50 seconds into the cycle. During the first 50 seconds, vertical bright streaks appear simultaneously across all frequency bands: the spectrogram signature of impulsive events, where a sharp time-domain spike produces instantaneous broadband energy. After 50 seconds, these vertical streaks are replaced by a sustained horizontal band at 58 Hz: the signature of continuous sinusoidal shaker excitation. The pressure panel confirms this transition coincides precisely with the depressurization of Side A (dropping from approximately 6,500 psi to 1,500 psi), demonstrating that the impulsive vibration is driven by the hydraulic actuation sequence, not the shaker.



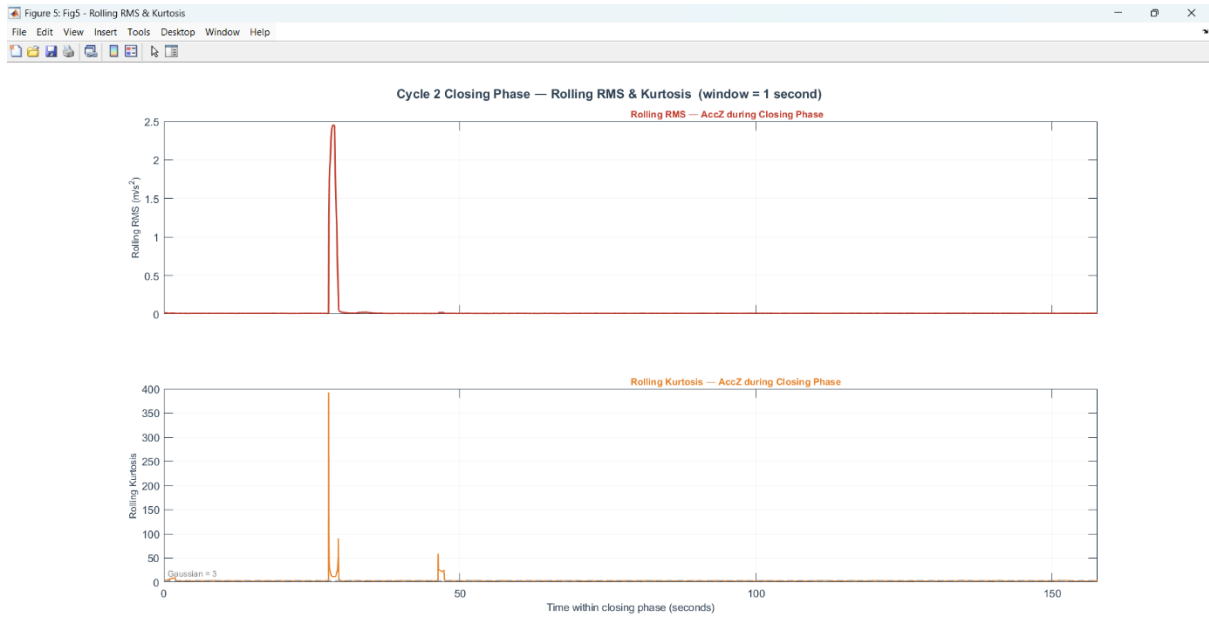
[Figure 4.7 — multi-parameter overview Cycle 1: MagZ position, AccX/Y/Z time series, AccX/Y/Z spectrograms, actuator pressure on shared time axis.]

### 4.3.6 Cycle 2 Closing Phase Anomaly

The anomalous Cycle 2 closing stroke was investigated in detail. Compared to a normal closing stroke (Cycle 1, approximately 8 seconds), Cycle 2 shows: closing duration 158 seconds; AccZ RMS 0.196 m/s<sup>2</sup> versus 0.006 m/s<sup>2</sup> (33 times higher); AccZ peak amplitude 11.58 m/s<sup>2</sup> versus 0.065 m/s<sup>2</sup> (178 times higher); AccZ kurtosis approximately 1,800 versus approximately 6 (300 times higher). The MagZ signal confirmed genuine slow and irregular gate travel during the closing stroke, with multiple partial contact events as the gate slowly descended under insufficient hydraulic pressure release rate. This is consistent with the engineer's assessment that the dump valve was not fully opened on this occasion, resulting in a slower-than-normal pressure release and repeated mechanical impacts as the gate intermittently contacted the seat surfaces.



[Figure 4.8 — Cycle 1 vs Cycle 2 AccZ comparison.]



*[Figure 4.9 —Rolling RMS/kurtosis through the Cycle 2 closing phase.]*

#### 4.4 Summary of Key Findings

Four principal findings emerge from this analysis. First, cycle-by-cycle KPI tracking successfully identifies the onset and progression of seat leakage failure (5k valve), distinguishes deliberate test escalation from degradation (10k valve), and confirms stable qualification performance (15k valve). Second, stroke duration KPIs across all 92 HF cycles reveal a systematic increase in both opening and closing times across test days, providing a potential indicator of hydraulic system state. Third, vibration analysis confirms a component bedding-in process in early cycles, visible as decreasing kurtosis and increasing RMS, with impulsive mechanical shocks transitioning to sustained broadband vibration. Fourth, multi-parameter visualization combining MagZ, accelerometer signals and pressure data on a shared time axis enables direct correlation of mechanical events with hydraulic conditions, providing substantially richer diagnostic information than any single channel alone.



## **Chapter 5: Conclusions and Future Work**

### **5.1 Introduction**

This chapter draws together the principal findings, reflects on their significance in relation to the research aim and objectives, discusses limitations, and proposes directions for future work. The chapter concludes with a statement on the potential for this work to transition into a sustained research collaboration.

### **5.2 Conclusions**

#### **5.2.1 Response to Research Objectives**

All five research objectives have been addressed. The first objective — analyzing gate valve actuation cycles using low-frequency time-series data — was met through KPI-based trend analysis across 500 cycles per valve class, successfully identifying distinct behavioural signatures across all three pressure classes. The second — extracting KPIs reflecting sealing quality and operational condition — was met through four primary indicators (peak hydraulic pressure, differential pressure, average and minimum flow rate) that proved diagnostically informative in all three valve datasets. The third — evaluating cycle repeatability and degradation trends — was met through rolling statistical analysis of the 500-cycle datasets and stroke duration KPI analysis across all 92 HF cycles. The fourth — analyzing high-frequency sensor data — was met through comprehensive RMS, kurtosis, PSD, and STFT analysis of cycles 1-5. The fifth — identifying characteristic signatures and anomalies — was met through identification of the 5k valve seat failure, the 10k test step-change, the Cycle 2 anomalous closing event, and the progressive stroke duration trend across all 92 HF cycles.

#### **5.2.2 Principal Findings**

The principal finding of the cyclic leakage test analysis is that cycle-by-cycle KPI tracking provides substantially richer diagnostic information than the binary pass/fail outcome of a qualification test. The 5k valve failure was detectable as a progressive trend before the final failure event, supporting Han et al. (2020), who argue that leakage should be treated as a quantitative diagnostic problem.

The principal finding of the HF vibration test analysis is that the combination of MagZ valve position, accelerometer vibration and actuator pressure on a shared time axis provides a multi-dimensional cycle signature substantially more informative than any single channel alone. The



transition from impulsive to broadband vibration across the first five cycles, the correlation of vibration events with hydraulic pressure phases, and the anomalous Cycle 2 closing behaviour were all interpretable only through the multi-parameter combined view.

The stroke duration trend analysis across all 92 cycles, applied using the EVENTS markers provided by the engineer, represents an additional original contribution. The systematic increase from Day 1 to Day 3 suggests that the hydraulic system state was changing over the course of the test, connecting directly to the valve signature health model concepts of crack time and stroke length as key performance indicators used in TechnipFMC's internal framework.

### **5.3 Limitations**

Several limitations should be acknowledged. First, detailed vibration analysis was restricted to five of the 92 available HF cycles due to the computational and time constraints associated with extracting data from the 2.8 GB raw file. Conclusions about the full 92-cycle progression must therefore be treated as preliminary. Second, the pressure data was recorded at approximately 1 Hz compared to 400 Hz for the accelerometer, meaning fine-scale pressure transients during actuation events are not resolved. Third, test operator logs were not available, so anomalous events could only be interpreted through post-hoc correspondence with the TechnipFMC engineer rather than through contemporaneous records. Fourth, the study is exploratory and does not include a formal validation step against independently confirmed fault states.

### **5.4 Future Work**

- Extension of HF vibration analysis to all 92 cycles: completing RMS, kurtosis, PSD and STFT analysis for the full dataset, with particular focus on the anomalous cycles C49, C66, C69, C73 and C74, to test whether the bedding-in trend observed in cycles 1-5 is maintained and to provide a robust Day 1 versus Day 3 comparison.
- Cross-referencing actuator pressure with MagZ for anomalous closing cycles: the TechnipFMC engineer specifically requested this analysis to determine whether slow closure is attributable to dump valve operation or other hydraulic factors.
- Investigating the air supply pressure hypothesis: comparing actuator pressure ramp-up rates between early and late cycles to test whether HPU output declined progressively over the three test days, consistent with the compressed air supply depletion hypothesis.

- Application of machine learning for anomaly classification: the structured KPI datasets generated in this study provide a natural input for unsupervised clustering or supervised classification methods to automate anomaly detection beyond the manual identification performed here.
- Integration with TechnipFMC's Valve Signature Health Model: the findings are directly relevant to TechnipFMC's internal framework for frac valve health scoring. Extending that framework to gate valves and incorporating the vibration dimension not currently present in the frac valve model represents a high-value continuation of this work.

## **5.5 Project Continuation and Mentoring**

This dissertation represents the initial exploratory phase of what could become a sustained research program. The datasets provided by TechnipFMC are richer than could be fully exploited within the scope of a single MSc project, and the analytical framework developed here provides a foundation for more advanced investigation. I am willing to mentor a future MSc or PhD student to continue and extend this analysis, building towards a validated multi-sensor condition monitoring methodology for subsea and surface gate valves in direct collaboration with TechnipFMC. Such a continuation would directly support TechnipFMC's stated strategic objective of developing proactive valve management capabilities and reducing unplanned maintenance costs.

## **5.6 Dissertation Summary**

This dissertation has developed and applied a data-driven exploratory analysis framework for the condition monitoring of industrial gate valves using experimental qualification test data from TechnipFMC. Cyclic leakage tests on 5k, 10k, and 15k pressure class valves were analyzed using a cycle-by-cycle KPI framework, identifying the onset of seat leakage failure, detecting test condition changes, and confirming stable performance across 500 cycles. High-frequency vibration test data from 92 actuation cycles were analyzed using MagZ-based valve position signals, stroke duration KPIs, and RMS, kurtosis, PSD, and STFT methods, revealing a component bedding-in process, a progressive hydraulic system trend, and an anomalous closing event attributable to manual operator technique.

The study demonstrates that integrating time-series KPI analysis with multi-parameter vibration characterization provides substantially richer diagnostic information than the binary pass/fail outcome of a conventional qualification test. The approach is consistent with the

direction of TechnipFMC's internal valve health modelling work and with the academic literature on data-driven condition monitoring. Future work should extend the framework to the full 92-cycle HF dataset, cross-reference pressure and vibration signatures for anomalous cycles, and explore automated classification methods to further advance the condition monitoring capability.

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## Appendix A: Data File Structure

The following table provides a complete inventory of all datasets, extracted files and MATLAB analysis scripts produced during this project. It is intended to support reproducibility and to assist any future researcher or engineer who wishes to extend this work.

| Root Folder                       | Sub-folder / File                                                    | Description                                                                                                                                                                                                |
|-----------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HF Limit Tests                    | A03 - Acc and Mag Data.txt                                           | Original 2.3 GB raw accelerometer and magnetometer file. 8 columns: DateTime, AccX, AccY, AccZ, MagX, MagY, MagZ. Recorded at 400 Hz (acc) and 100 Hz (mag). Full 3-day HF test.                           |
| HF Limit Tests                    | A03 - Acc and Mag Data with cycle markers.txt                        | Updated 2.8 GB file from TechnipFMC engineer. Identical to original with additional EVENTS column marking open/close cycle boundaries for all 92 cycles.                                                   |
| HF Limit Tests                    | 1775311830804_A03_-_Pressure_Data_only.csv                           | Pressure data at ~1 Hz. 4 columns: DateTime (MM:SS.s), Actuator (psi), Side A (psi), Side B (psi). Covers full 4.6-hour test duration.                                                                     |
| HF Limit Tests \ Extracted Cycles | Cycle_01.csv – Cycle_05.csv                                          | Individual cycle files extracted using PowerShell. Each file covers one complete actuation cycle (open start to close end + 15s padding). Format matches main acc/mag file.                                |
| HF Limit Tests \ Extracted Cycles | Cycle_49.csv, Cycle_66.csv, Cycle_69.csv, Cycle_73.csv, Cycle_74.csv | Anomalous cycles identified from stroke duration KPI analysis. Extracted for detailed investigation of abnormal closing behaviour (closing durations 100–195s vs typical 30–80s).                          |
| HF Limit Tests                    | EVENTS_ONLY.xlsx                                                     | Excel file extracted using Power Query. 370 rows, 2 columns: DateTime and EVENTS. Precise timestamps for all 92 complete cycle events across 3 test days.                                                  |
| Cyclic Leakage Tests \ 5k Valve   | *.csv (10 files)                                                     | Raw cyclic leakage test data for the 5,000 psi valve class. 50 cycles per file, 500 cycles total. Columns: DateTime, HydPressure, Flowmeter, DiffPressure. Test result: FAIL (seat leakage at cycle ~430). |
| Cyclic Leakage Tests \ 10k Valve  | *.csv (10 files)                                                     | Raw cyclic leakage test data for 10,000 psi valve class. 500 cycles total. Same                                                                                                                            |

| Root Folder                      | Sub-folder / File               | Description                                                                                                                                                                                                    |
|----------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  |                                 | column structure as 5k valve dataset.<br>Test result: PASS.                                                                                                                                                    |
| Cyclic Leakage Tests \ 15k Valve | *.csv (10 files)                | Raw cyclic leakage test data for 15,000 psi valve class. 500 cycles total. Additional columns include bore pressure, cavity pressure and temperature. Test result: PASS.                                       |
| MATLAB Scripts                   | hf_engineers_meeting_analysis.m | Combined analysis script addressing all 6 points from TechnipFMC engineering meeting. Produces 6 figures: MagZ overlay, per-cycle 5-panel, opening overlay, RMS chart, David signature plot, labelled scatter. |
| MATLAB Scripts                   | hf_stroke_kpi_analysis.m        | Stroke duration KPI analysis for all 92 cycles using EVENTS timestamps. Produces 4 figures including trend plots with rolling mean and summary statistics table.                                               |
| MATLAB Scripts                   | hf_cycle2_anomaly_analysis.m    | Detailed investigation of the Cycle 2 anomalous closing stroke. Produces 5 figures including STFT spectrogram and rolling RMS/kurtosis through the 158s closing phase.                                         |
| MATLAB Scripts                   | gate_valve_plots_500cycles.m    | KPI trend analysis for 5k valve cyclic leakage test (500 cycles). Produces 4-panel KPI trend plots with rolling statistics and anomaly zoom.                                                                   |
| MATLAB Scripts                   | gate_valve_plots_10k.m          | KPI trend analysis for the 10k valve cyclic leakage test.                                                                                                                                                      |
| MATLAB Scripts                   | gate_valve_15k_plots.m          | KPI trend analysis for the 15k valve cyclic leakage test.                                                                                                                                                      |

## Appendix B: Skills and Competencies

The table below maps the key skills developed and applied during this project to their descriptions and relevant MSc Applied Data Science in Engineering (APDSCE) modules, following the competency framework used in the program.

| Skill                | Description                                                                                                                                                                            | Relevant APDSCE Module                                    |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Domain Knowledge     | Working knowledge of gate valve operation, hydraulic actuation, API 6A qualification testing and subsea oil and gas systems gained through industry collaboration with TechnipFMC      | PdM, Data Capture, System Health Management               |
| Data Handling        | Processing large raw sensor files (2.8 GB accelerometer file, 500-cycle pressure datasets) using PowerShell extraction, MATLAB preprocessing pipelines and Python validation           | Software Development for Data Science                     |
| Signal Processing    | Implementation of RMS, Kurtosis, PSD (Welch method), STFT, and Symmetrized Dot Pattern analysis from first principles without specialist toolboxes                                     | Data Visualization, System Health Management              |
| Programming Skills   | Full MATLAB and Python analysis pipelines for data extraction, cycle segmentation, KPI extraction, feature computation and multi-panel visualization across multiple datasets          | Data Visualization, Software Development for Data Science |
| Communication Skills | Presenting findings to academic supervisors and industry engineers at TechnipFMC, interpreting technical feedback and responding with targeted analytical results                      | —                                                         |
| Strategy             | Developing a systematic data-driven analysis framework combining low-frequency KPI monitoring with high-frequency vibration analysis, prioritizing analysis based on engineer feedback | System Health Management, Data Capture, PdM               |
| Condition Monitoring | Applying industry-standard and novel diagnostic methods (SDP, STFT, Kurtosis) to real qualification test data to identify degradation trends, anomalies and valve condition indicators | PdM, System Health Management                             |

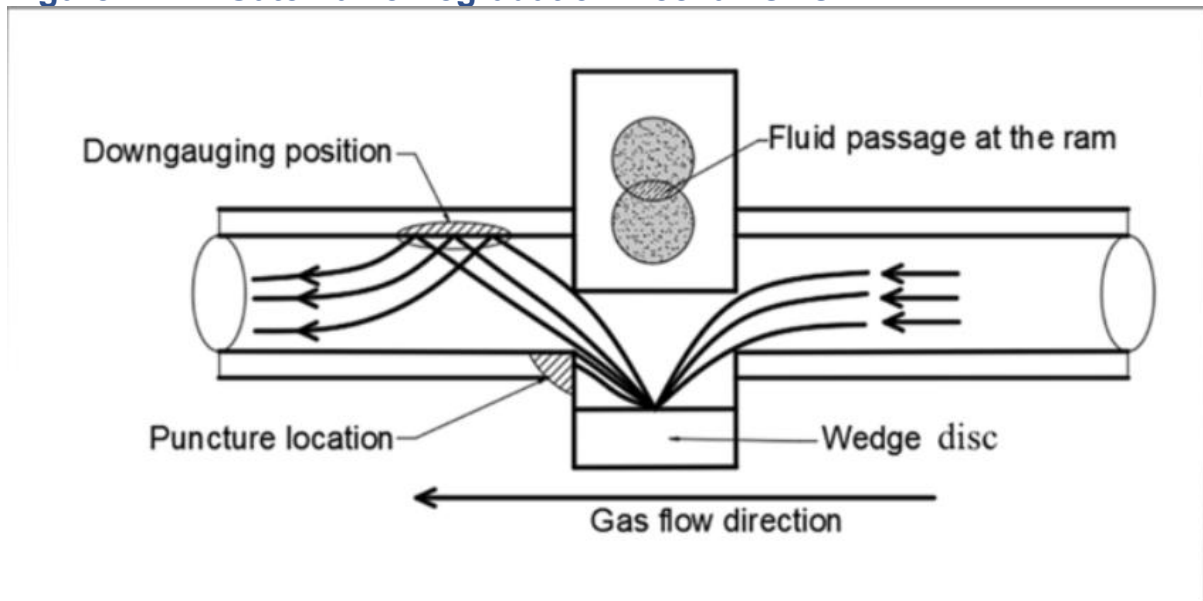


This project was conducted in collaboration with TechnipFMC and supervised by Dr Octavian Niculita (Glasgow Caledonian University). The author acknowledges the guidance and technical support provided by **Ben Alderson**, **David Faichnie**, and **Peter Wallace** of TechnipFMC throughout the project.

## Appendix C: Figures and Explanation

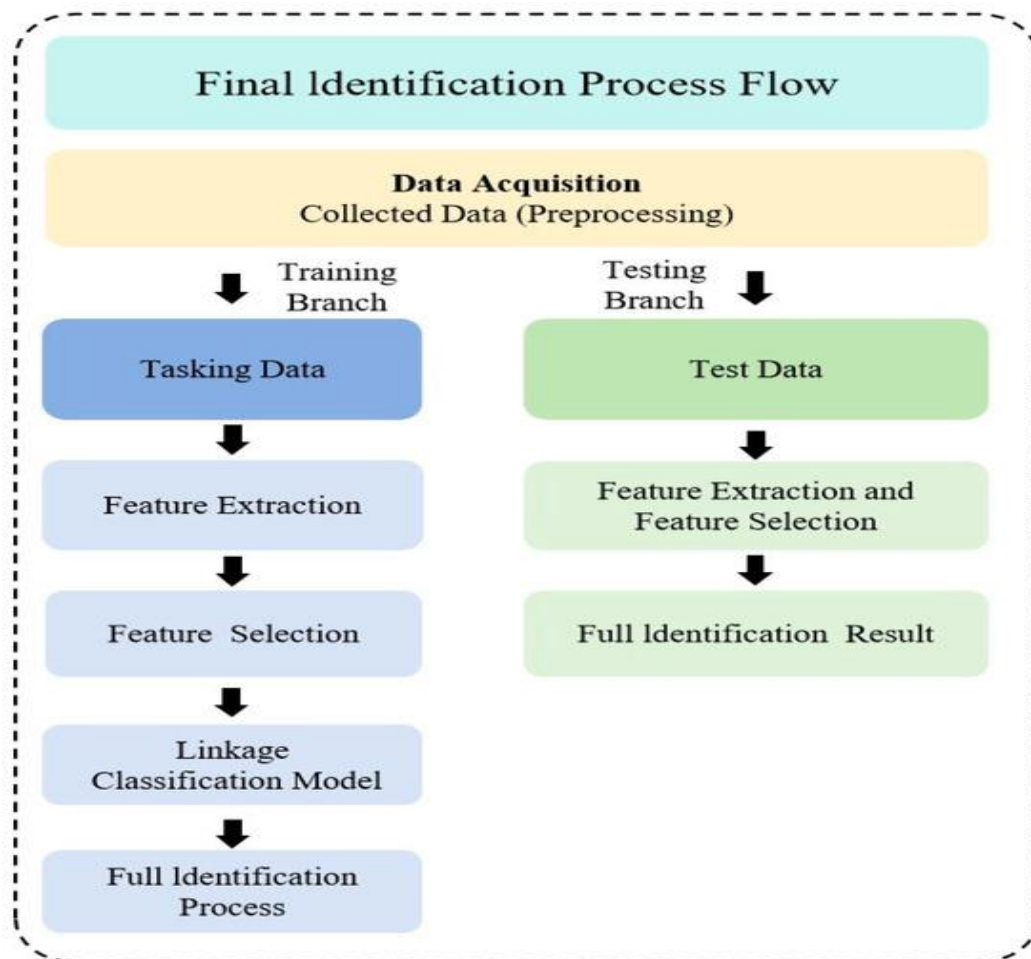
### Chapter 2 Figures — Literature Review

Figure 2.1 — Gate Valve Degradation Mechanisms



The above diagram describes the internal degradation of a gate valve during its functioning. The first part of the closing assembly is the wedge disc, and as the gas moves from right to left through the device, it will move along the fluid channel within the ram. The fast-flowing gas causes the erosion at the puncture point – an area at the bottom of the valve that is vulnerable since the fluid concentrates there. This process of downgauging is seen at the downgauging point of the diagram. In essence, the picture presents the gradual wear of the valve's sealing surface as a result of constant cycling and flow.

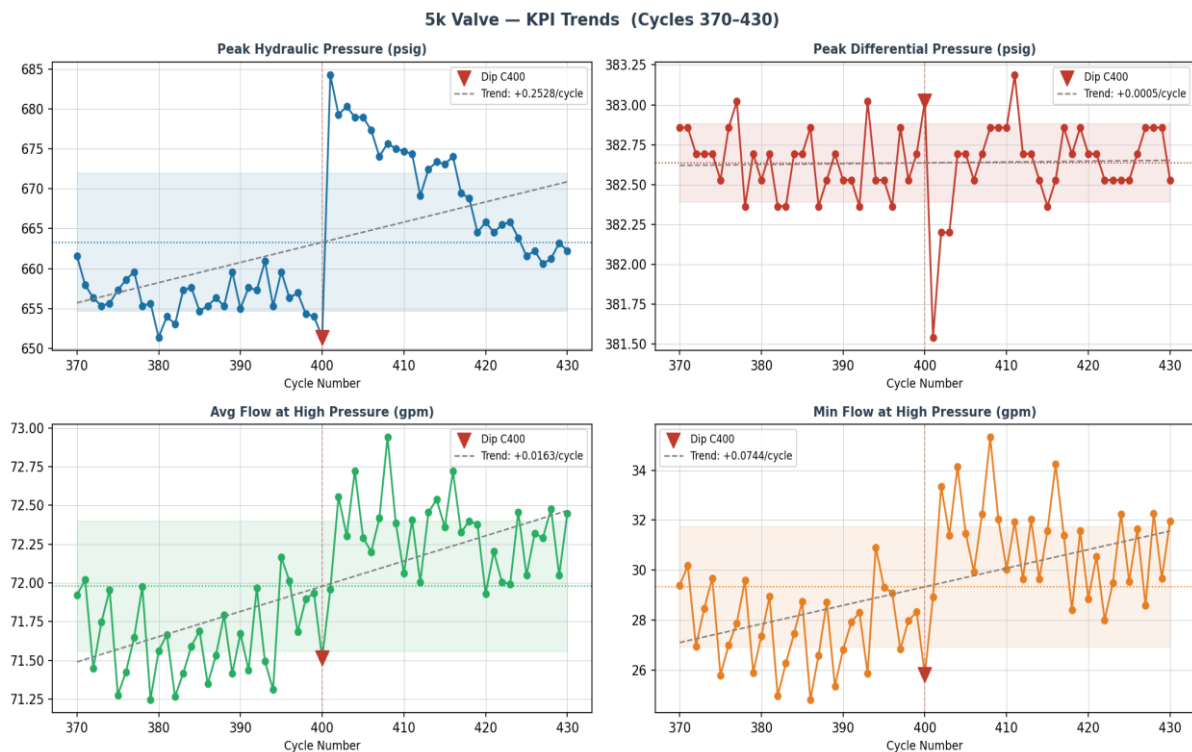
**Figure 2.2 — Condition Monitoring Framework for Gate Valve Performance**



The above diagram shows the structure of the machine learning process that allows for determining the condition of the valve. The first step of the process is collecting data. Then, the entire process splits into two parallel flows. The first one is related to constructing the identification system based on past data (tasking data). To do so, the system should go through feature extraction, feature selection, and the linkage classification stage. The second flow deals with analyzing new data from the valve (the data obtained from Ben's files) via the same stages (feature extraction and selection). As a result, you obtain identification information about the valve's condition.

## Chapter 4 Figures — Results and Analysis

Figure 4.1 — 5k Valve KPI Trends, Cycles 370–430



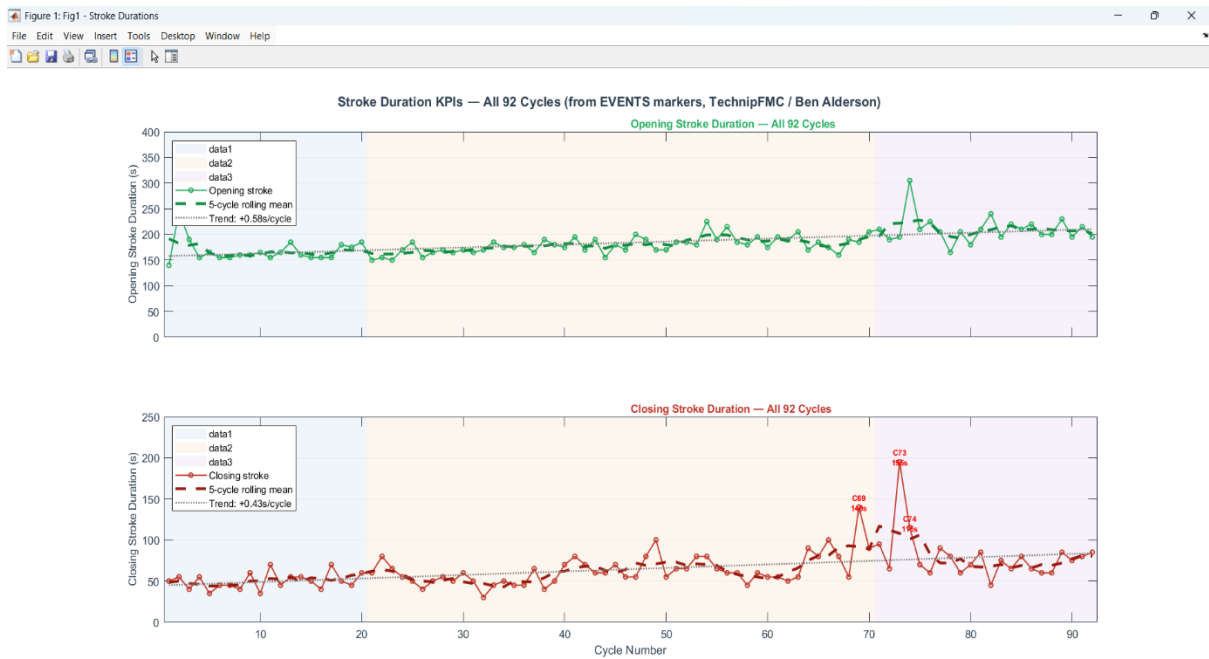
The above graph gives us the Cycle-by-cycle KPI trend plots for the 5,000 psi gate valve, zoomed to cycles 370–430 where the degradation signature is most pronounced. Four KPIs are tracked across this window: **peak hydraulic pressure (psig)**, **peak differential pressure (psid)**, **average flow rate** during the hold phase (gpm), and **minimum flow rate** during the hold phase (gpm). The 25-cycle rolling mean and  $\pm 1$  standard deviation band are overlaid. The critical finding is the progressive collapse of hydraulic pressure beginning at approximately cycle 408, followed by the complete loss of differential pressure by cycle 430, confirming that the valve seat can no longer maintain a seal under the applied load. The minimum flow rate simultaneously rises towards the maximum measurable value, indicating significant leakage through the closed seat. A brief anomalous pressure dip visible at approximately cycle 400 may represent an early-warning precursor signal occurring approximately 8 cycles before the onset of visible degradation. This figure was presented to TechnipFMC to support the test-fail determination and to demonstrate that the KPI framework can identify the onset of degradation before catastrophic leakage occurs.

## Figure 4.2 — 10k Valve KPI Trends, Cycles 44–56



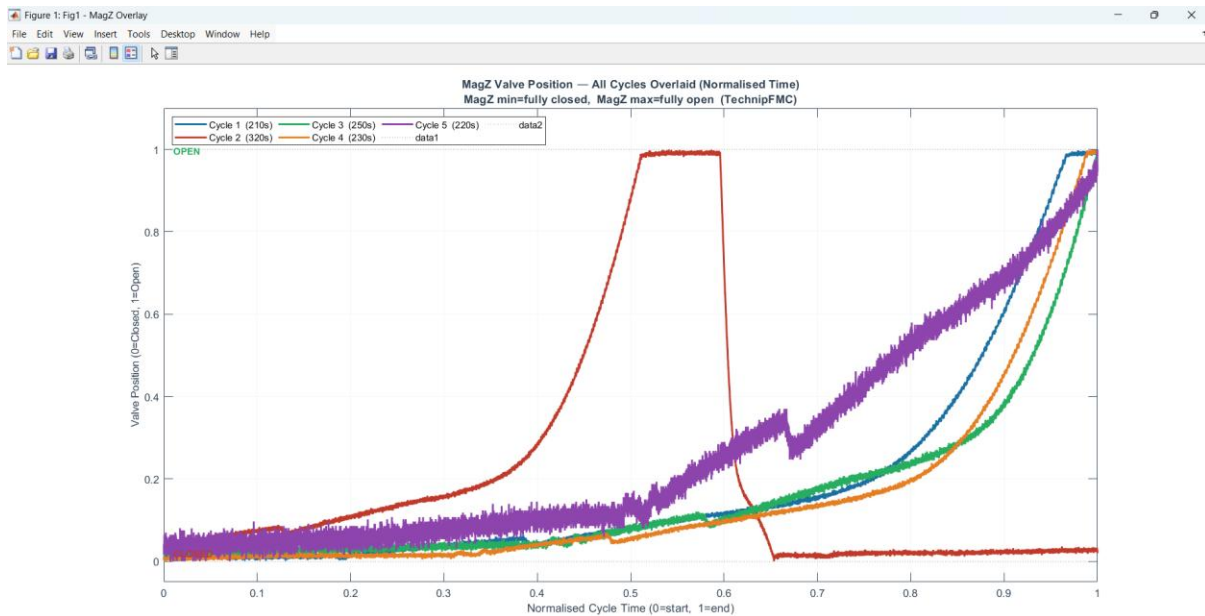
Zoomed KPI view of the 10,000 psi gate valve test, covering cycles 44–56. This window captures the most notable feature in the 10k valve dataset: an anomalous period of near-zero minimum flow rate at cycles 47–54, followed by a step increase in hydraulic pressure at cycle 55. The near-zero leakage window indicates the valve was sealing almost perfectly under the lower test pressure in force at that time. The pressure step at cycle 55 represents a deliberate increase in test severity applied by the TechnipFMC test engineers — this is a known test condition change, not a valve fault. After the pressure step, the KPIs stabilize at the new, higher level and remain within acceptance criteria for the remainder of the 500-cycle test. This figure was produced to demonstrate that the KPI framework correctly distinguishes between genuine valve anomalies and planned test condition changes, which is a critical capability for a condition monitoring system used in a qualification test context.

## Figure 4.3 — Stroke Duration KPIs Across All 92 Cycles



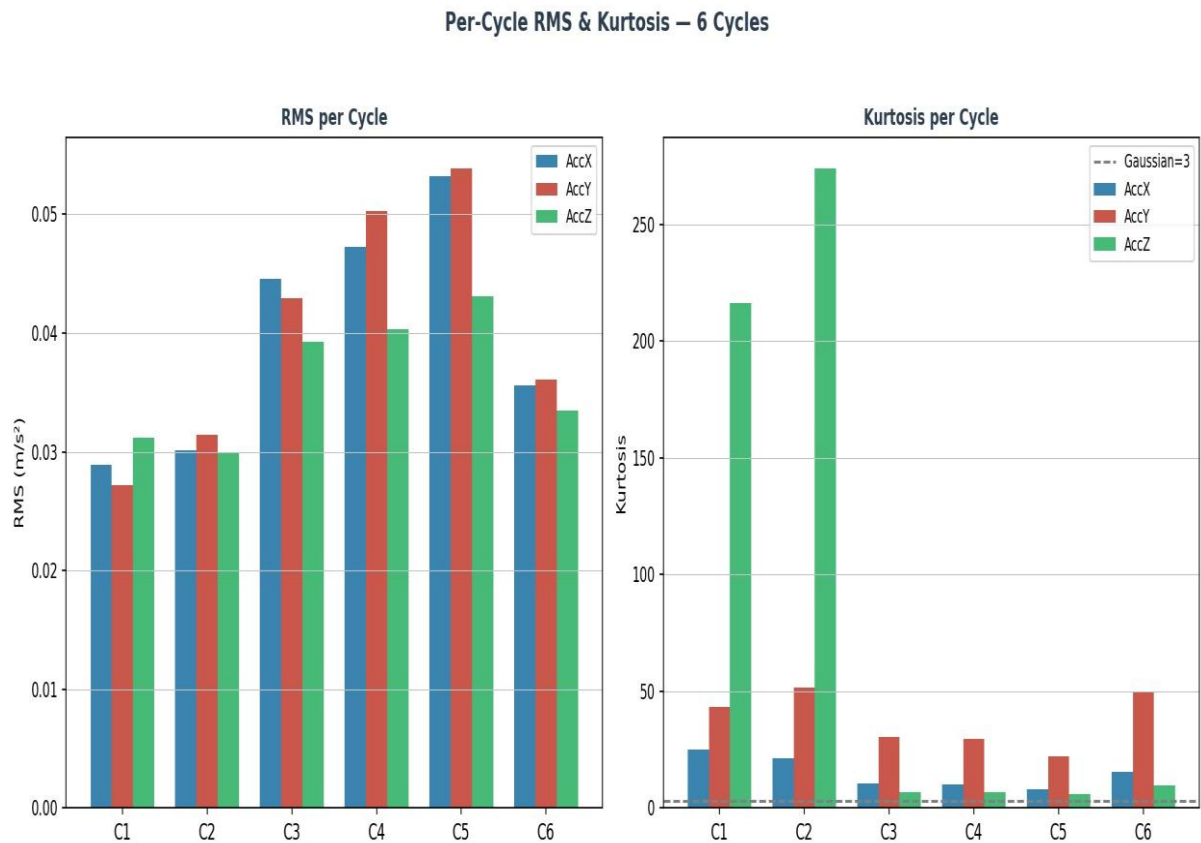
In this plot, the number of cycles is on the X-axis, while the time is on the Y-axis to illustrate how long each cycle's opening and closing periods are. The opening period is shown by the green graph, while the closing period is shown by the red graph. It can be noted that both graphs have an increasing trend, namely,  $+0.58$  s/cycle for the opening period and  $+0.43$  s/cycle for the closing period. Hence, it can be concluded that the valve operation is becoming increasingly difficult and might have experienced some mechanical wear. The opening period duration varies between 170 s and 220 s, whereas the closing period ranges from 50 s to 60 s. However, there is a significant increase in the closing period from cycles 69 to 74, especially for C69, C73, and C74.

## Figure 4.4 — MagZ Normalized Overlay for Cycles 1–5



This graph depicts the normalized measurements of the MagZ sensors for 5 separate cycles where each cycle has been normalized so that it starts at time zero (when fully closed) and ends at time one (when fully open). What makes this interesting is the fact that there is a large variation among cycles as far as getting to the point when the valve is opened completely. The blue, green, and orange cycles follow a similar pattern of slowly increasing until close to time one when fully open. On the other hand, red shows a very unusual behavior for cycle 2 where it increases to full open position at around time 0.4 and then decreases to nearly zero. It seems to indicate either a spurious signal or an error in data acquisition or an actual mechanical reversal.

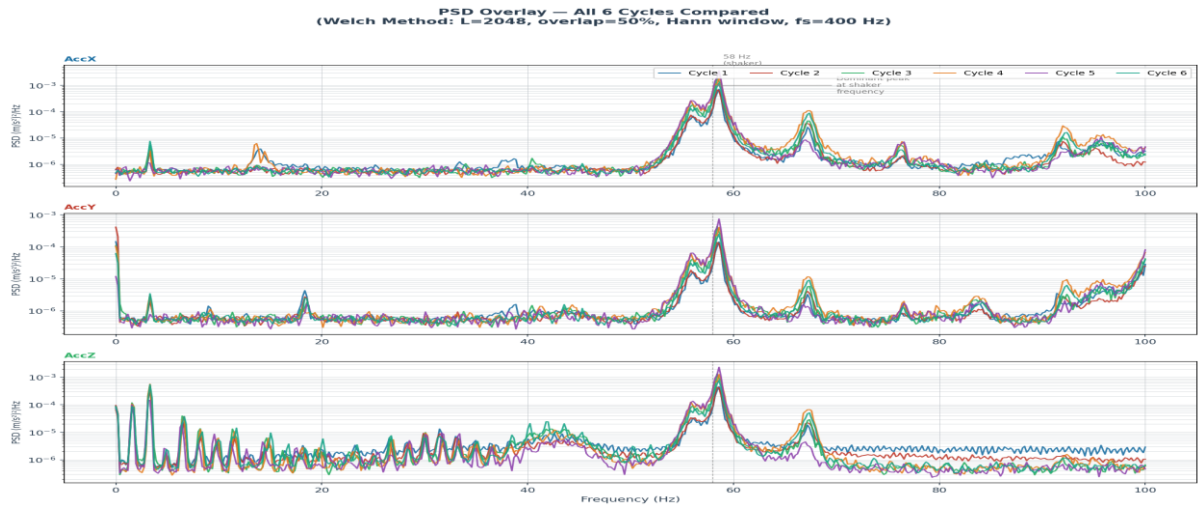
**Figure 4.5 — Per-Cycle RMS and Kurtosis for Cycles 1–6**



Bar chart comparison of RMS (Root Mean Square, left panel) and Kurtosis (right panel) computed for AccX, AccY and AccZ across six analysed cycles of the HF vibration test. RMS quantifies signal energy or average vibration magnitude; higher RMS indicates more vigorous dynamic activity. Kurtosis quantifies impulsiveness — the degree to which a signal is dominated by isolated large spikes rather than a smooth sustained waveform. For a perfectly Gaussian signal, kurtosis equals 3. The complementary diagnostic value of these two metrics is the key finding of this figure. Cycle 1 has relatively low RMS but extremely high AccZ kurtosis of 216, and Cycle 2 has the highest kurtosis of all six cycles at 274 — both far above the Gaussian baseline of 3. This indicates highly impulsive, spike-dominated vibration consistent with repeated mechanical impacts during a bedding-in phase, as confirmed by the TechnipFMC engineers. By Cycles 3–6, kurtosis drops sharply to approximately 6–10 while RMS increases, indicating a transition to sustained broadband vibration characteristic of settled valve operation. Using RMS alone would have led to the incorrect conclusion that early cycles are the least significant; it is kurtosis that reveals the mechanically important impulsive character of the initial actuations.

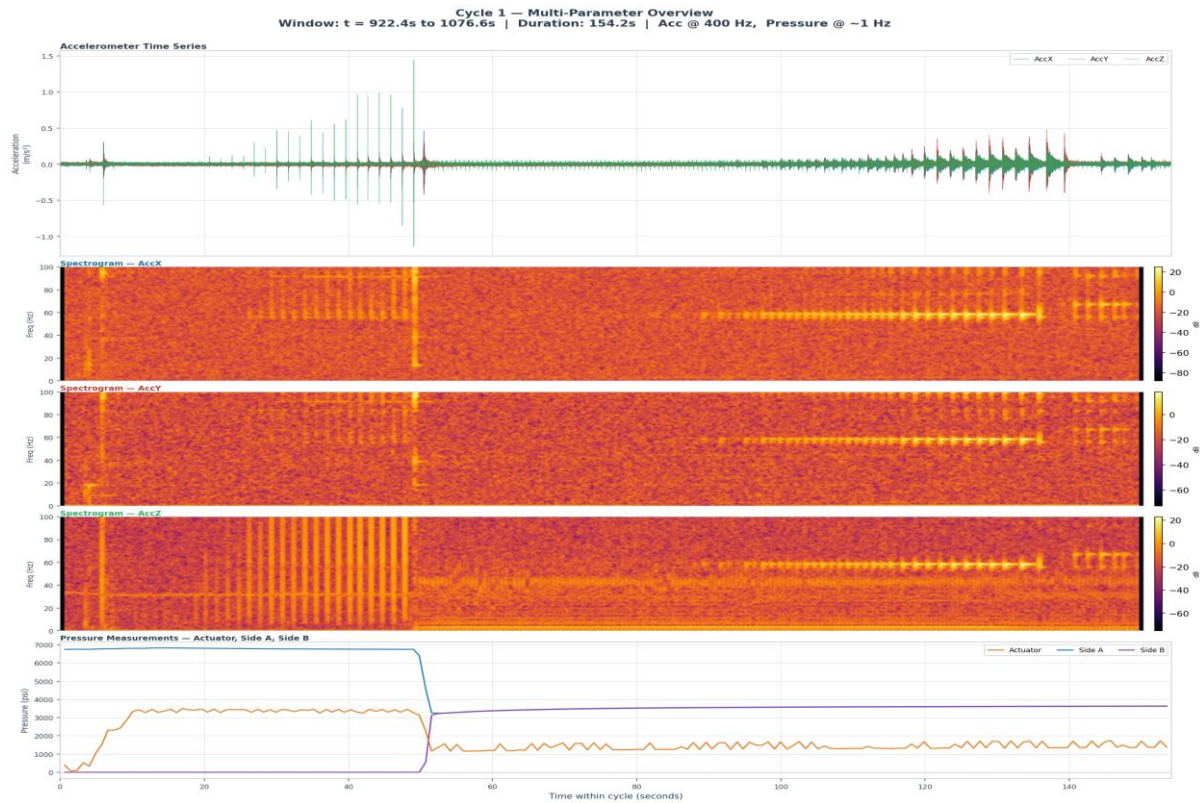


**Figure 4.6 — PSD Overlay Showing Dominant 58 Hz Peak, Cycles 1–5**



Power Spectral Density (PSD) overlay for cycles 1–5 of the HF vibration test, computed separately for AccX (top panel), AccY (middle panel), and AccZ (bottom panel). PSD was estimated using Welch's method implemented from first principles: 2048-sample Hann-windowed segments with 50% overlap, averaged across all segments within each cycle. The frequency resolution is  $400/2048 = 0.195$  Hz, and the analysis is restricted to 0–100 Hz. The dominant peak at 58 Hz, annotated with a dashed vertical line, is present in all cycles and all axes and corresponds to the shaker excitation frequency applied to the valve body throughout the HF test. The most diagnostically informative feature is the elevated low-frequency energy (0–20 Hz region) visible in cycles 1 and 2 compared to cycles 3–5. This low-frequency enrichment is characteristic of impulsive signals: sparse high-amplitude transient events in the time domain produce broadband energy concentrated at low frequencies in the spectral domain. This observation is fully consistent with the kurtosis findings in Figure 4.5 and provides spectral confirmation of the bedding-in vibration regime in the first two cycles.

**Figure 4.7 — Multi-Parameter Overview: Cycle 1**



Multi-parameter overview of Cycle 1 of the HF vibration test, produced for a combined view of all sensor signals on a shared time axis. The figure contains seven panels: the accelerometer time series (AccX, AccY, AccZ overlaid), three STFT spectrograms (AccX, AccY, AccZ individually), and the actuator pressure signal from the separate pressure data acquisition system. All panels share the same time axis, enabling direct visual correlation between valve position (MagZ, shown as an overlay on the time series panel), vibration behavior, frequency content, and hydraulic pressure at every moment of the cycle. The STFT spectrograms (512-sample Hann window, 480-sample overlap) show the 58 Hz shaker excitation as a horizontal bright stripe and the impulsive bedding-in events as isolated vertical bright streaks during the opening phase of Cycle 1.

## Figure 4.8 — Cycle 1 vs Cycle 2: Closing Phase AccZ Comparison

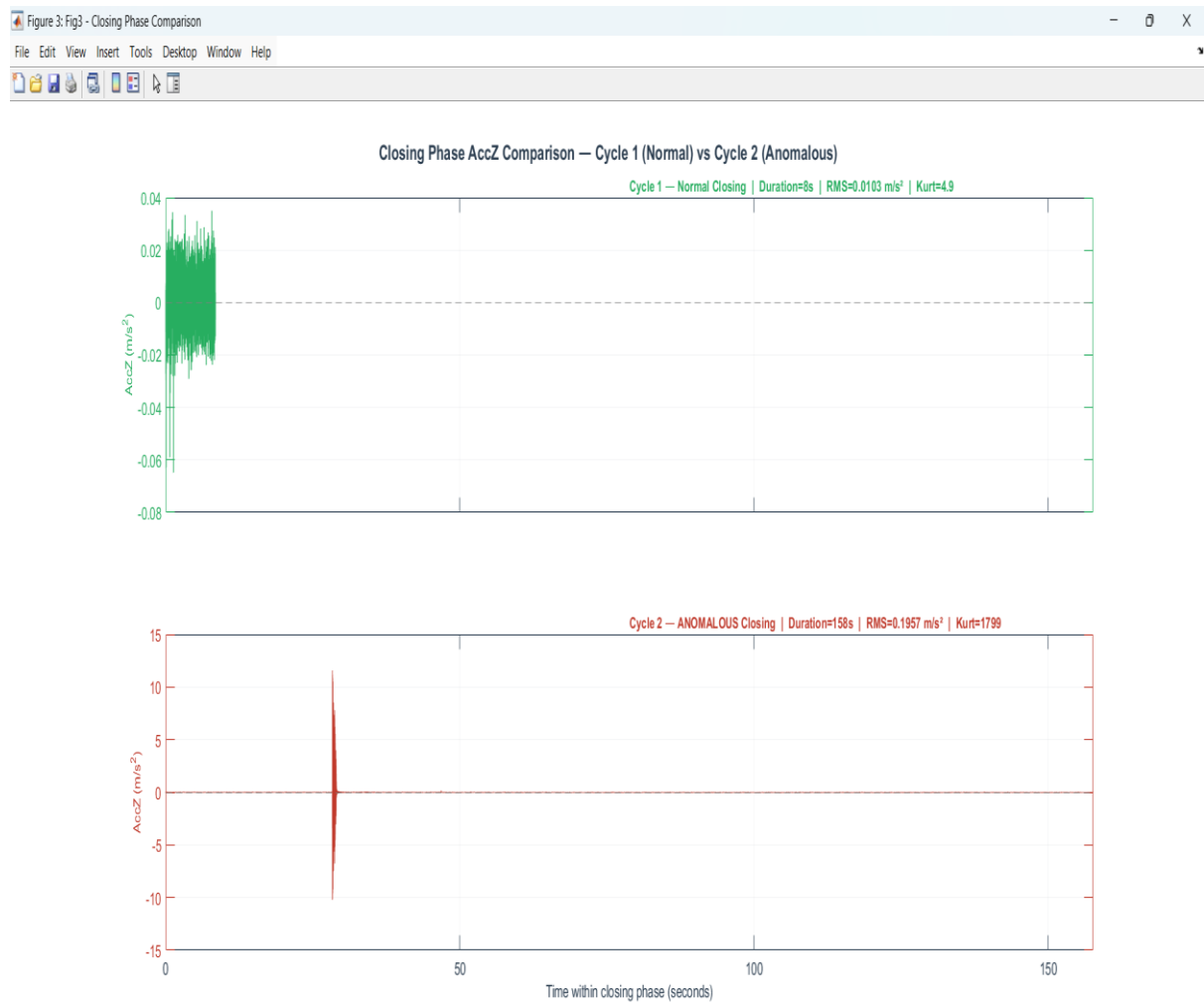
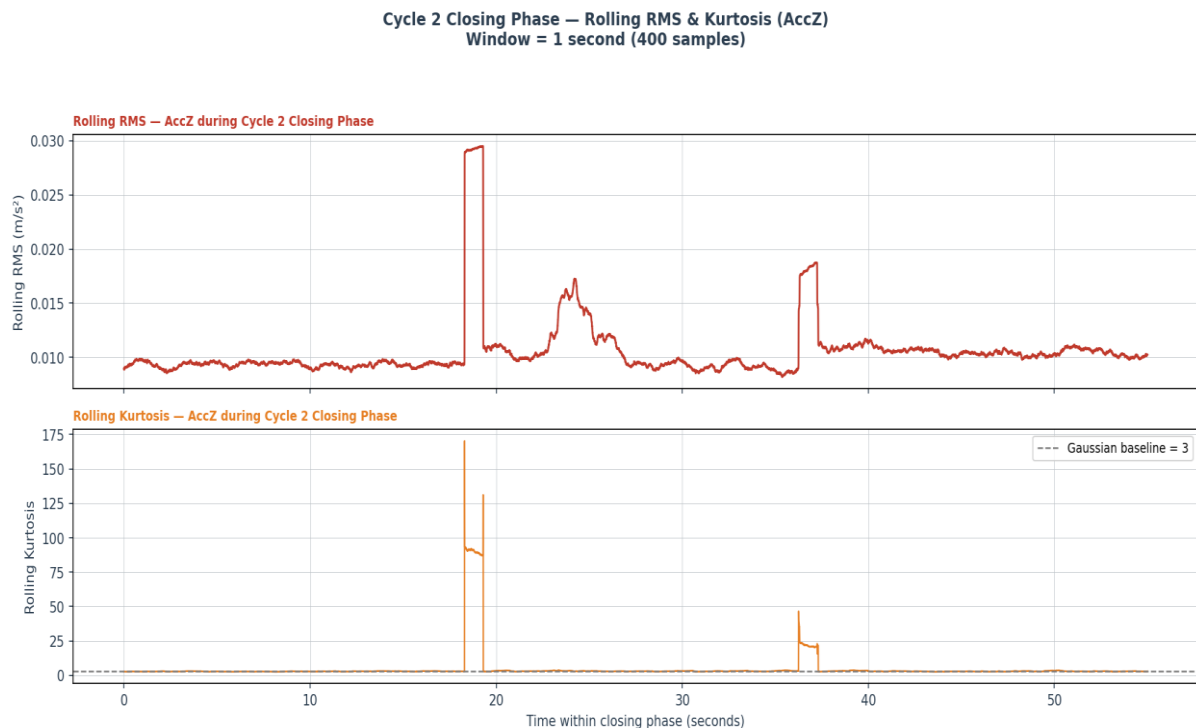


Figure 1 shows the AccZ sensor output during the closing phase for two cycles. For cycle 1 (shown by green lines on top), the output is normal; the vibration activity is confined to just about the beginning of the closing phase (first 10 seconds) with an RMS value of 0.0103 m/s<sup>2</sup> and a Kurtosis value of 4.9, after which the output goes down to close to zero. This is typical of a normal operation; there is mechanical activity while closing and none thereafter.

For cycle 2 (red lines on bottom), however, the situation is vastly different. The cycle is marked as anomalous, as the closing phase is extremely long, lasting 158 seconds against 8 seconds for cycle 1. There is a very strong impulse of about  $\pm 10$  m/s<sup>2</sup> around the 30-second mark. The RMS is also 0.1957 m/s<sup>2</sup> (nearly 20 times larger than the previous one), and Kurtosis is 1799, whereas normal mechanical signals typically lie below 10.

**Figure 4.9 — Rolling RMS and Kurtosis Through Cycle 2 Closing Phase**



This graph magnifies the unusual closing phase of Cycle 2 by using rolling statistics over a 1-second window, thereby pinpointing the point where the impact occurs. From the graph, the Rolling RMS (upper plot, red line) produces a sudden spike at the approximate value of 2.4 m/s<sup>2</sup> within one second after 30 seconds, after which the value drops back to almost zero, proving that the impact is indeed a single occurrence of intense energy rather than continuous vibration.

The Rolling Kurtosis (lower plot, orange line) provides even more concrete evidence about this issue; it also gives a sudden spike at almost 400 at the same time, followed by another smaller spike close to 45 seconds, with the Gaussian reference value of 3 plotted for comparison. This extremely high value is characteristic of an impulsive fault in rotating and reciprocating equipment. Thus, from these two graphs, it becomes evident that the impact occurred at this very instant.